



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Konfigurasi Vlan Trunk OSPF Multivendor

Rahmat Maulana Ansori - 5024241011

2026

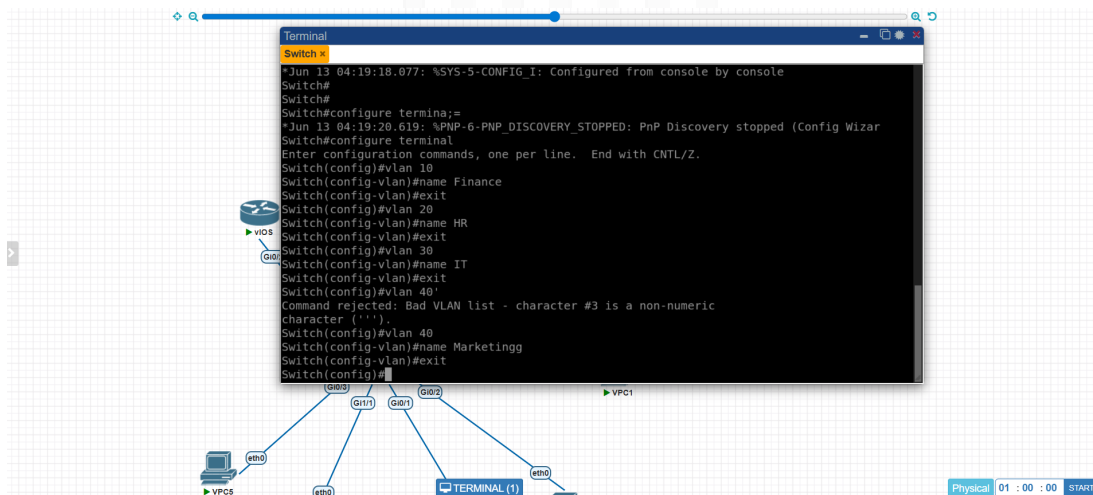
1 Langkah-Langkah Percobaan

Praktikum Modul 5 terdiri atas lima tahapan utama, yaitu konfigurasi Cisco Switch, konfigurasi Cisco Router, konfigurasi Huawei Router beserta pengalamatan IP antarrouter, konfigurasi routing dinamis OSPF multivendor, dan pengujian failover OSPF. Seluruh percobaan dijalankan pada simulator jaringan PNETLab.

1.1 Praktikum 1: Konfigurasi Cisco Switch

Praktikum pertama mencakup pembuatan VLAN, pengaturan *access port* ke setiap PC, dan pengaturan *trunk port* menuju Cisco Router pada Cisco Switch. Saat pertama kali booting, dipilih opsi **no** pada dialog konfigurasi awal, kemudian ditekan Enter. Konfigurasi disimpan menggunakan perintah `write memory` setelah seluruh tahap selesai.

Pertama-tama, dilakukan akses ke mode konfigurasi global pada Cisco Switch. Kemudian dibuat empat VLAN, yaitu VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing), di mana setiap VLAN diberi nama yang mencerminkan fungsi departemen yang diwakilinya.



Gambar 1: Pembuatan VLAN 10, 20, 30, dan 40

Setelah pembuatan VLAN selesai, dijalankan perintah `show vlan brief` untuk memverifikasi bahwa keempat VLAN telah terdaftar dan berstatus *active*.

```
terminal
Switch x
Switch(config-vlan)#exit
Switch(config)#show vlan brief
^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#configure termina;
*Jun 13 04:21:08.221: %SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
                                           Gi1/0, Gi1/1, Gi1/2, Gi1/3
10   Finance                 active
20   HR                     active
30   IT                     active
40   Marketingg             active
1002 fddi-default          act/unsup
1003 token-ring-default   act/unsup
1004 fddinet-default      act/unsup
1005 trnet-default        act/unsup
Switch#
```

Gambar 2: Output perintah `show vlan brief` pada Cisco Switch

Selanjutnya, setiap interface yang menghubungkan PC dikonfigurasi sebagai *access port* dan dialokasikan ke VLAN yang sesuai. Nomor interface disesuaikan dengan topologi yang digunakan.

```
Terminal
Switch x
1004 fddinet-default          act/unsup
1005 trnet-default            act/unsup
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface gi0/2
Switch(config-if)#description PC-FINANCE-VLAN10
Switch(config-if)#switchport mode access
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/1
Switch(config-if)#description PC-HR-VLAN20
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan20
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/2
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#

% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/2
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#

Terminal
Switch x
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/2
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface gi0/1
Switch(config-if)#description PC-IT-VLAN30
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan30
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/3
Switch(config-if)#description PC-MARKETING-VLAN40
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

Gambar 3: Konfigurasi *access port* untuk setiap VLAN

Interface yang menghubungkan Cisco Router kemudian dikonfigurasi sebagai *trunk port* dengan enkapsulasi 802.1Q, sehingga VLAN 10, 20, 30, dan 40 dapat melintas secara bersamaan melalui satu link fisik. Rincian konfigurasi trunk ditunjukkan pada tabel berikut.

Tabel 1: Konfigurasi *trunk port* pada Cisco Switch

Interface Switch	Terhubung ke	Mode	VLAN Diizinkan
Gi0/0	Router Cisco Gi0/2	Trunk	10, 20, 30, 40

Setelah konfigurasi trunk selesai, dijalankan perintah `show interfaces trunk` untuk memverifikasi bahwa *trunk port* aktif dan membawa keempat VLAN.


```
Terminal
Switch x
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#description TRUNK-T0-CISCO-ROUTER
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#sh
*Jun 13 04:30:12.592: %SYS-5-CONFIG_I: Configured from console by conso
Switch#show interface trunk

Port      Mode          Encapsulation  Status        Native vlan
Gi0/0     on            802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
```

Gambar 4: Output perintah show interfaces trunk pada Cisco Switch

Terakhir, seluruh konfigurasi disimpan menggunakan perintah write memory.

```
Terminal
Switch x
Switch#sh
*Jun 13 04:30:12.592: %SYS-5-CONFIG_I: Configured from console by conso
Switch#show interface trunk

Port      Mode          Encapsulation  Status        Native vlan
Gi0/0     on            802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#write memory
Building configuration...
Compressed configuration from 3365 bytes to 1646 bytes[OK]
Switch#
*Jun 13 04:30:43.152: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk.
Please wait...
*Jun 13 04:30:43.820: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk succes
sfully.
```

Gambar 5: Penyimpanan seluruh konfigurasi Cisco Switch

1.2 Praktikum 2: Konfigurasi Cisco Router

Praktikum kedua mencakup konfigurasi Cisco Router sebagai *gateway* untuk keempat VLAN menggunakan metode ROAS (*Router on a Stick*) melalui *subinterface*, serta konfigurasi DHCP server untuk VLAN 10 dan VLAN 20. Saat pertama kali booting, dipilih opsi `no` pada dialog konfigurasi awal. Konfigurasi disimpan menggunakan perintah `do wr` di dalam mode konfigurasi, atau `wr` di luar mode konfigurasi.

Pertama-tama, dilakukan akses ke mode konfigurasi global pada Cisco Router.


```
terminal
Switch x Mikrotik x vIOS x
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/2
Router(config-if)#description TRUNK-TO-CISCO-SWITCH
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 13 04:37:00.126: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
*Jun 13 04:37:01.126: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, c
hanged state to up
Router(config)#interface gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.20
Router(config-subif)#description GATEWAY-VLAN20-HR
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

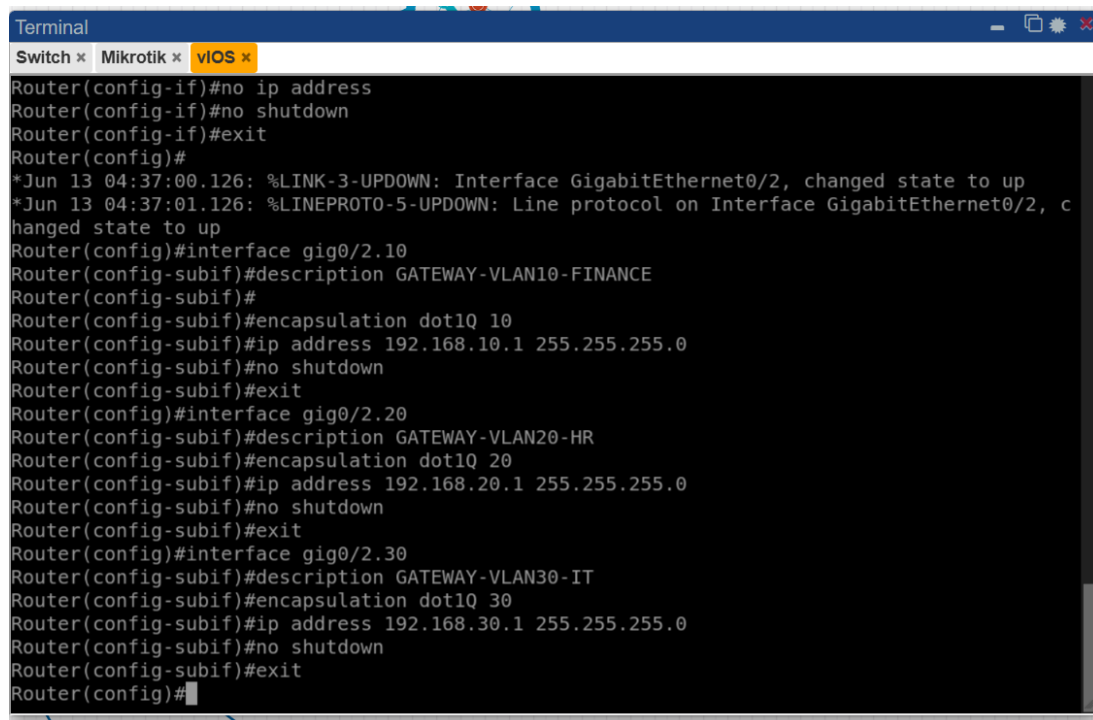
Gambar 8: Konfigurasi *subinterface* gig0/2.10 untuk VLAN 10

Subinterface gig0/2.20 dikonfigurasi untuk VLAN 20 dengan IP address 192.168.20.1/24.

```
terminal
Switch x Mikrotik x vIOS x
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/2
Router(config-if)#description TRUNK-TO-CISCO-SWITCH
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 13 04:37:00.126: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
*Jun 13 04:37:01.126: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, c
hanged state to up
Router(config)#interface gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.20
Router(config-subif)#description GATEWAY-VLAN20-HR
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

Gambar 9: Konfigurasi *subinterface* gig0/2.20 untuk VLAN 20

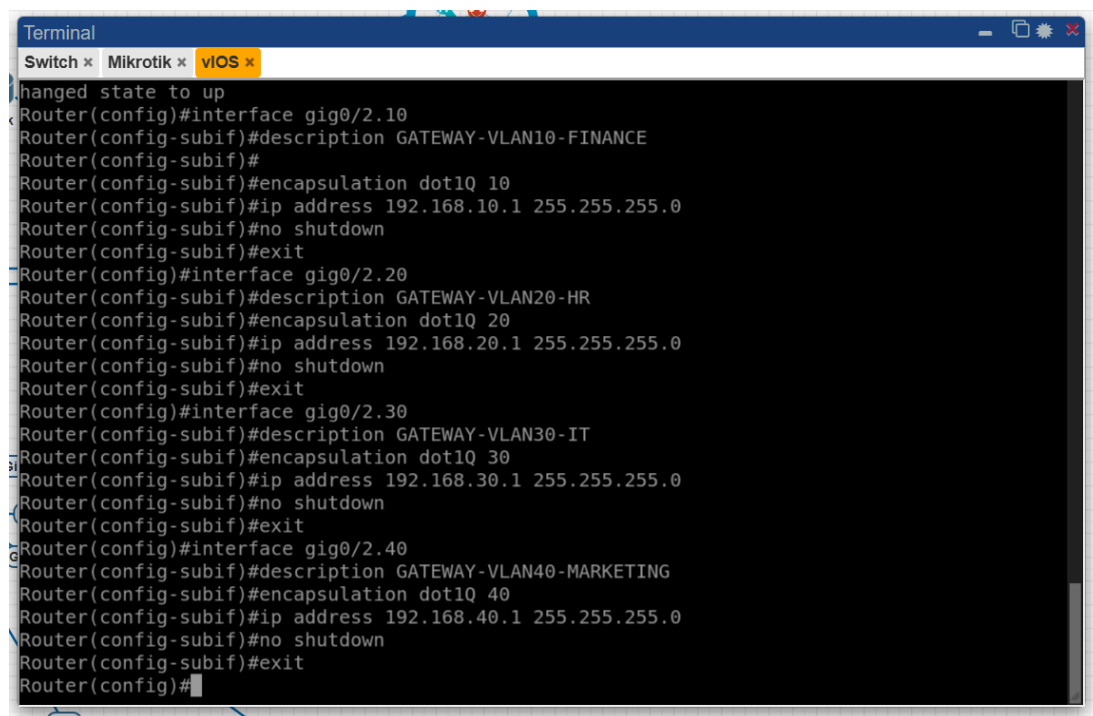
Subinterface gig0/2.30 dikonfigurasi untuk VLAN 30 dengan IP address 192.168.30.1/24.

A terminal window titled 'Terminal' with tabs for 'Switch', 'Mikrotik', and 'vIOS'. The 'vIOS' tab is active. The terminal shows the configuration of subinterface gig0/2.30. It starts with 'Router(config-if)#no ip address', 'Router(config-if)#no shutdown', and 'Router(config-if)#exit'. Then it enters 'Router(config)#'. There are two status messages: '*Jun 13 04:37:00.126: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up' and '*Jun 13 04:37:01.126: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up'. Then it configures 'interface gig0/2.10' with description 'GATEWAY-VLAN10-FINANCE', encapsulation 'dot1Q 10', and IP address '192.168.10.1 255.255.255.0'. Next, it configures 'interface gig0/2.20' with description 'GATEWAY-VLAN20-HR', encapsulation 'dot1Q 20', and IP address '192.168.20.1 255.255.255.0'. Finally, it configures 'interface gig0/2.30' with description 'GATEWAY-VLAN30-IT', encapsulation 'dot1Q 30', and IP address '192.168.30.1 255.255.255.0'. The prompt is 'Router(config)#'.

```
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 13 04:37:00.126: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
*Jun 13 04:37:01.126: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, c
changed state to up
Router(config)#interface gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.20
Router(config-subif)#description GATEWAY-VLAN20-HR
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.30
Router(config-subif)#description GATEWAY-VLAN30-IT
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

Gambar 10: Konfigurasi *subinterface* gig0/2.30 untuk VLAN 30

Subinterface gig0/2.40 dikonfigurasi untuk VLAN 40 dengan IP address 192.168.40.1/24.

A terminal window titled 'Terminal' with tabs for 'Switch', 'Mikrotik', and 'vIOS'. The 'vIOS' tab is active. The terminal shows the configuration of subinterface gig0/2.40. It starts with 'Router(config)#interface gig0/2.10' with description 'GATEWAY-VLAN10-FINANCE', encapsulation 'dot1Q 10', and IP address '192.168.10.1 255.255.255.0'. Then it configures 'interface gig0/2.20' with description 'GATEWAY-VLAN20-HR', encapsulation 'dot1Q 20', and IP address '192.168.20.1 255.255.255.0'. Next, it configures 'interface gig0/2.30' with description 'GATEWAY-VLAN30-IT', encapsulation 'dot1Q 30', and IP address '192.168.30.1 255.255.255.0'. Finally, it configures 'interface gig0/2.40' with description 'GATEWAY-VLAN40-MARKETING', encapsulation 'dot1Q 40', and IP address '192.168.40.1 255.255.255.0'. The prompt is 'Router(config)#'.

```
Router(config)#interface gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.20
Router(config-subif)#description GATEWAY-VLAN20-HR
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.30
Router(config-subif)#description GATEWAY-VLAN30-IT
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.40
Router(config-subif)#description GATEWAY-VLAN40-MARKETING
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

Gambar 11: Konfigurasi *subinterface* gig0/2.40 untuk VLAN 40

Selanjutnya dikonfigurasi DHCP server untuk VLAN 10. Rentang IP 192.168.10.1 hingga 192.168.10.10 dikecualikan dari pool agar tidak didistribusikan kepada klien.

```

Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#

```

Gambar 12: Konfigurasi DHCP server untuk VLAN 10

DHCP server untuk VLAN 20 juga dikonfigurasi dengan cara yang sama, dengan rentang IP 192.168.20.1 hingga 192.168.20.10 dikecualikan dari pool.

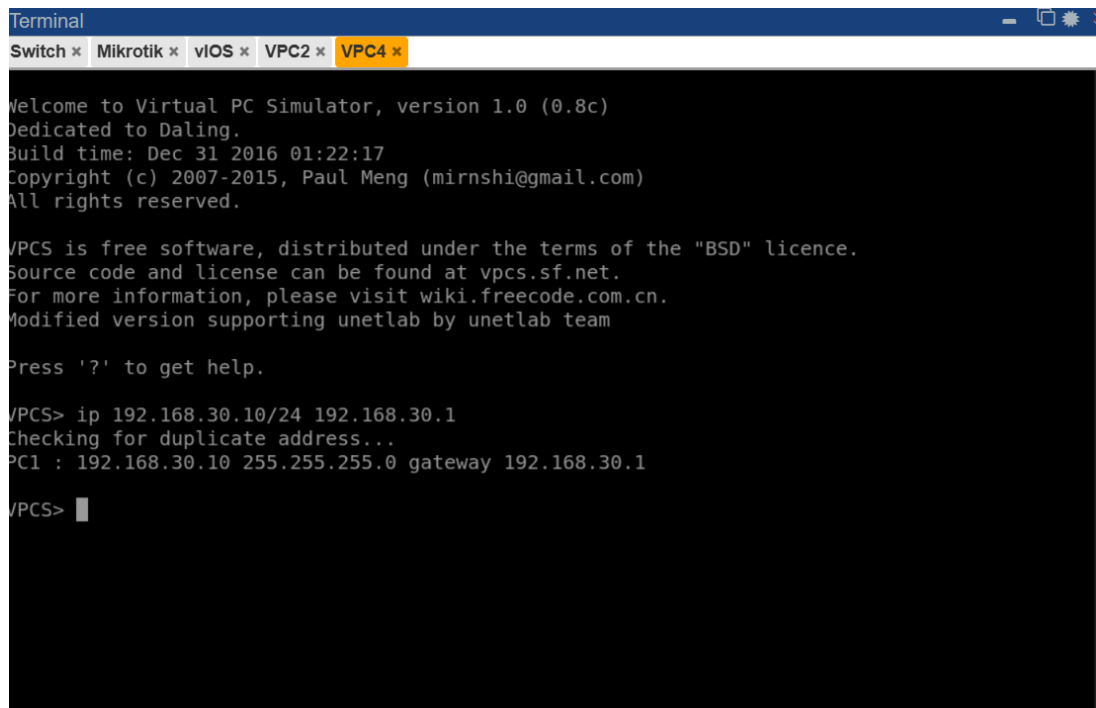
```

Terminal
Switch x Mikrotik x vIOS x
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.30
Router(config-subif)#description GATEWAY-VLAN30-IT
Router(config-subif)#encapsulation dot1q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.40
Router(config-subif)#description GATEWAY-VLAN40-MARKETING
Router(config-subif)#encapsulation dot1q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#

```

Gambar 13: Konfigurasi DHCP server untuk VLAN 20

Untuk VLAN 30 dan VLAN 40, pengalamatan IP dilakukan secara statis langsung pada masing-masing VPCS. PC VLAN 30 dikonfigurasi dengan IP 192.168.30.10/24 dan *gateway* 192.168.30.1, sedangkan PC VLAN 40 dikonfigurasi dengan IP 192.168.40.10/24 dan *gateway* 192.168.40.1.



```
Terminal
Switch x Mikrotik x vIOS x VPC2 x VPC4 x
Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

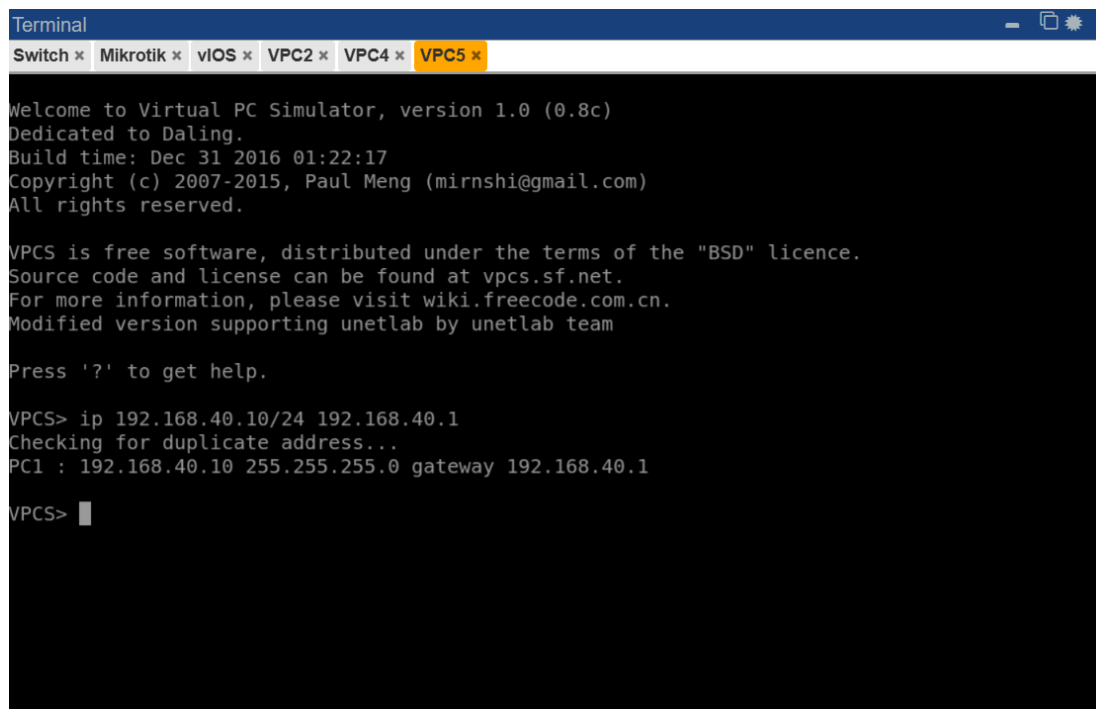
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS>
```

Gambar 14: Konfigurasi IP statis pada PC VLAN 30



```
Terminal
Switch x Mikrotik x vIOS x VPC2 x VPC4 x VPC5 x
Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

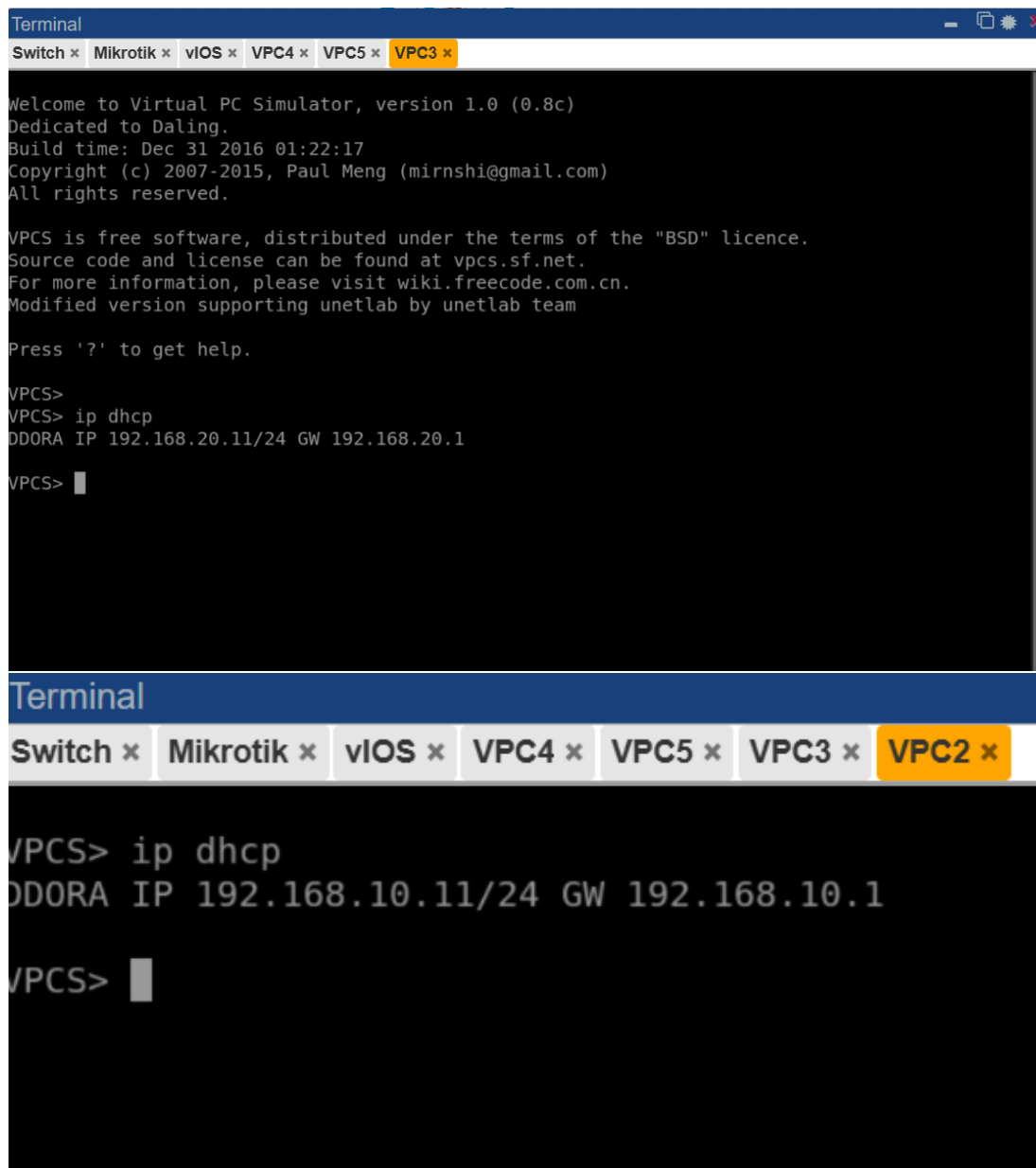
Press '?' to get help.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS>
```

Gambar 15: Konfigurasi IP statis pada PC VLAN 40

Sementara itu, VPCS pada VLAN 10 dan VLAN 20 dikonfigurasi untuk memperoleh IP secara otomatis melalui DHCP.



Gambar 16: VPCS VLAN 10 dan VLAN 20 memperoleh IP melalui DHCP

Verifikasi dilakukan dengan menjalankan perintah `show ip interface brief` untuk memastikan seluruh *subinterface* aktif dengan IP address yang benar.

```
Terminal
Switch x Mikrotik x VIOS x VPC4 x VPC5 x VPC3 x VPC2 x
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip 192.168.30.10/24 192.168.30.1
^
% Invalid input detected at '^' marker.

Router(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#show verison
*Jun 13 04:52:27.852: %SYS-5-CONFIG_I: Configured from console
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES unset    administratively down down
GigabitEthernet0/1       unassigned      YES unset    administratively down down
GigabitEthernet0/2       unassigned      YES unset    up          up
GigabitEthernet0/2.10    192.168.10.1    YES manual   up          up
GigabitEthernet0/2.20    192.168.20.1    YES manual   up          up
GigabitEthernet0/2.30    192.168.30.1    YES manual   up          up
GigabitEthernet0/2.40    192.168.40.1    YES manual   up          up
GigabitEthernet0/3       unassigned      YES unset    administratively down down
Router#
```

Gambar 17: Output show ip interface brief pada Cisco Router

Perintah show ip dhcp pool juga dijalankan untuk memastikan pool DHCP telah terbentuk dengan konfigurasi yang tepat.


```
Terminal
Switch * Mikrotik * vIOS * VPC4 * VPC5 * VPC3 * VPC2 *
GigabitEthernet0/3      unassigned      YES unset administratively down down
Router#show ip dhcp pull
^
% Invalid input detected at '^' marker.
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses               : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  192.168.10.12      192.168.10.1 - 192.168.10.254      1

Pool VLAN20-HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses               : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  192.168.20.12      192.168.20.1 - 192.168.20.254      1
Router#

Terminal
Switch * Mikrotik * vIOS * VPC4 * VPC5 * VPC3 * VPC2 *
Pool VLAN10-FINANCE :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses               : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  192.168.10.12      192.168.10.1 - 192.168.10.254      1

Pool VLAN20-HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses               : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  192.168.20.12      192.168.20.1 - 192.168.20.254      1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.d9    Jun 14 2026 04:51 AM    Automatic
192.168.20.11    0100.5079.6668.da    Jun 14 2026 04:51 AM    Automatic
Router#
```

Gambar 18: Output show ip dhcp pool pada Cisco Router

Pengujian perolehan IP dan konektivitas ke *gateway* dilakukan pada setiap VLAN. PC VLAN 10 meminta IP melalui DHCP dan berhasil melakukan ping ke *gateway* 192.168.10.1.

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x

VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86395, 86400/43200/75600
MAC        : 00:50:79:66:68:d9
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> 
```

Gambar 19: PC VLAN 10 berhasil memperoleh IP dari DHCP

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86395, 86400/43200/75600
MAC        : 00:50:79:66:68:d9
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.847 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.995 ms
^C
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.790 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.231 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.220 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=1.848 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=1.645 ms

VPCS> 
```

Gambar 20: Ping dari PC VLAN 10 ke *gateway* 192.168.10.1

PC VLAN 20 juga meminta IP melalui DHCP dan berhasil melakukan ping ke *gateway* 192.168.20.1.

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86217, 86400/43200/75600
MAC        : 00:50:79:66:68:da
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS>
```

Gambar 21: PC VLAN 20 berhasil memperoleh IP dari DHCP

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x
VPCS>
VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86217, 86400/43200/75600
MAC        : 00:50:79:66:68:da
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.329 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=2.150 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.099 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.685 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.900 ms

VPCS>
```

Gambar 22: Ping dari PC VLAN 20 ke gateway 192.168.20.1

PC VLAN 30 dan PC VLAN 40 juga berhasil melakukan ping ke *gateway* masing-masing.

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.196 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.775 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.657 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=2.134 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=2.252 ms

VPCS>
```

Gambar 23: Ping dari PC VLAN 30 ke *gateway* 192.168.30.1

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.671 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.144 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=2.163 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=6.583 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=2.376 ms

VPCS>
```

Gambar 24: Ping dari PC VLAN 40 ke *gateway* 192.168.40.1

Pengujian ping antar-VLAN dilakukan dari PC VLAN 10 ke PC di VLAN 20, VLAN 30, dan VLAN 40 untuk memverifikasi bahwa *inter-VLAN routing* melalui Cisco Router berjalan dengan benar.

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=1.645 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=7.161 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=3.134 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=2.950 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=3.137 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=4.860 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=10.603 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=2.565 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.430 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.001 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=3.507 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=4.081 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.511 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.133 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=3.008 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.383 ms

VPCS>

```

Gambar 25: Ping antar-VLAN dari PC VLAN 10

Terakhir, perintah `show ip dhcp binding` dijalankan pada Cisco Router untuk memverifikasi daftar IP yang telah didistribusikan kepada klien. IP yang tercatat harus sesuai dengan alamat yang diterima oleh PC VLAN 10 dan VLAN 20.

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x
Current index      IP address range      Leased addresses
192.168.10.12      192.168.10.1 - 192.168.10.254  1

Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses             : 1
Pending event                : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.20.12      192.168.20.1 - 192.168.20.254  1

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/             Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.d9      Jun 14 2026 04:51 AM  Automatic
192.168.20.11    0100.5079.6668.da      Jun 14 2026 04:51 AM  Automatic
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/             Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.d9      Jun 14 2026 04:53 AM  Automatic
192.168.20.11    0100.5079.6668.da      Jun 14 2026 04:51 AM  Automatic
Router#

```

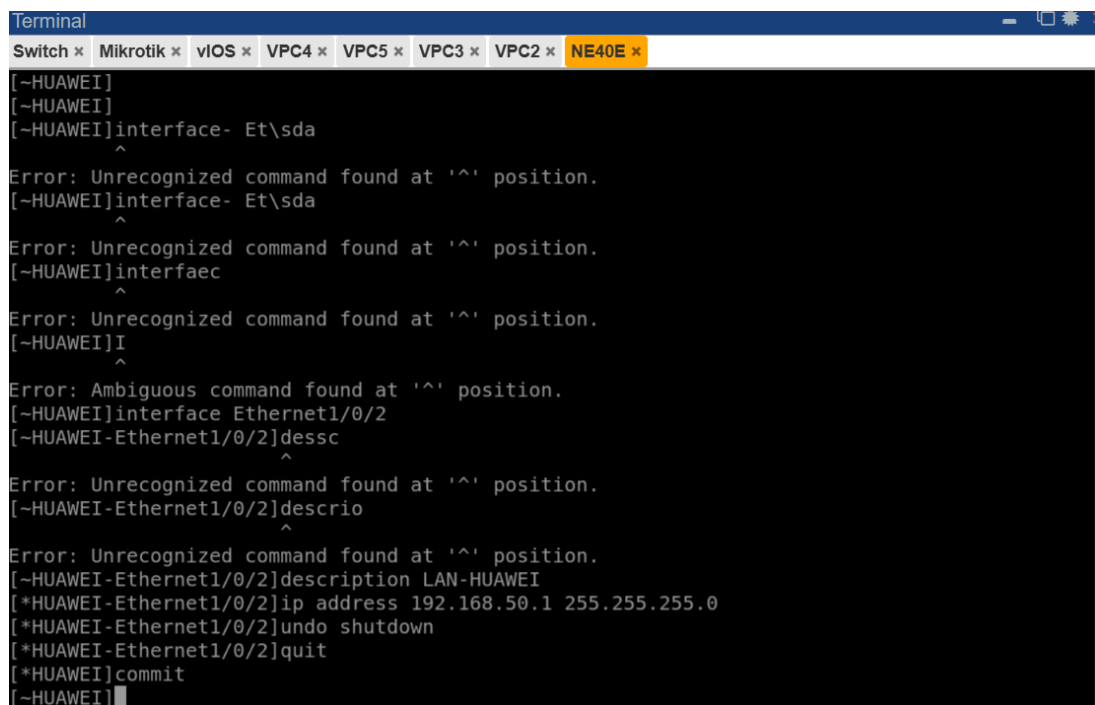
Gambar 26: Output `show ip dhcp binding` pada Cisco Router

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

Praktikum ketiga mencakup konfigurasi LAN Huawei dan pengalamatan IP pada seluruh link antarrouter, meliputi dua link Cisco ke Huawei, satu link Huawei ke MikroTik, dan satu link Cisco ke MikroTik. Setelah konfigurasi selesai, konektivitas langsung antarrouter diverifikasi sebelum OSPF dikonfigurasi.

Beberapa hal yang perlu diperhatikan: gunakan kombinasi tombol `Ctrl + H` pada terminal Huawei sebagai pengganti Backspace. Apabila saat booting Huawei Router masuk ke `root@debian:/opt/vsrp#`, matikan router, pilih opsi *wipe* melalui klik kanan, kemudian nyalakan kembali. Konfigurasi Huawei dianggap tersimpan apabila indikator terminal berubah dari tanda bintang menjadi tanda tilde; perintah `commit` dijalankan setelah setiap blok konfigurasi. Terdapat pula koreksi pada topologi, di mana `g0/3` terhubung ke `Ethernet1/0/0` dan `g0/0` terhubung ke `Ethernet1/0/4`.

Interface `Ethernet1/0/2` pada Huawei Router dikonfigurasi sebagai antarmuka LAN dengan jaringan `192.168.50.0/24` dan IP address `192.168.50.1`.



```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]interface- Et\sda
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]interface- Et\sda
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]interfaec
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]I
^
Error: Ambiguous command found at '^' position.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2]dessc
^
Error: Unrecognized command found at '^' position.
[~HUAWEI-Ethernet1/0/2]descrio
^
Error: Unrecognized command found at '^' position.
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 27: Konfigurasi LAN Huawei

PC yang terhubung ke LAN Huawei dikonfigurasi dengan IP statis `192.168.50.10/24` dan *gateway* `192.168.50.1`. Setelah itu, dijalankan perintah `show ip` untuk memverifikasi alamat yang telah dikonfigurasi.

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:d8
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:d8
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS>
```

Gambar 28: IP address VPCS LAN Huawei

Konektivitas ke *gateway* Huawei diuji melalui ping ke 192.168.50.1 untuk memastikan LAN berjalan dengan baik.

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
RHOST:PORT : 127.0.0.1:30000
MTU         : 1500

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:d8
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=10.112 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.120 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.446 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=1.742 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=1.917 ms

VPCS>
```

Gambar 29: Ping dari VPCS LAN Huawei ke *gateway* 192.168.50.1

Berikutnya dikonfigurasi link pertama antara Cisco dan Huawei menggunakan jaringan 10.10.10.0/30. Cisco Router menggunakan interface Gi0/3 dengan IP 10.10.10.1/30, sedangkan Huawei Router menggunakan interface Ethernet1/0/0 dengan IP 10.10.10.2/30.

Tabel 2: Pengalamatan link Cisco ke Huawei (Link 1)

Perangkat	Interface	IP Address
Cisco Router	Gi0/3	10.10.10.1/30
Huawei Router	Ethernet1/0/0	10.10.10.2/30

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Hardware address/
User name
92.168.10.11 0100.5079.6668.d9 Jun 14 2026 04:53 AM Automatic
92.168.20.11 0100.5079.6668.da Jun 14 2026 04:51 AM Automatic
router# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
router(config)#interface gi0/3 description LINK1-T0-HUAWEI
^
% Invalid input detected at '^' marker.

router(config)#interface gi0/3
router(config-if)#description LINK1-T0-HUAWEI
router(config-if)#ip address 10.10.10.1 255.255.255.252
router(config-if)#no shutdown
router(config-if)#exhit
^
% Invalid input detected at '^' marker.

router(config-if)#exhit
Jun 13 05:08:46.421: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
Jun 13 05:08:47.421: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, c
changed state no shutdown
router#
Jun 13 05:08:49.543: %SYS-5-CONFIG_I: Configured from console by console
router#
router#
router#
```

Gambar 30: Konfigurasi Link 1 pada Cisco Router

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
[~HUAWEI-Ethernet1/0/2]dessc
^
Error: Unrecognized command found at '^' position.
[~HUAWEI-Ethernet1/0/2]descrio
^
Error: Unrecognized command found at '^' position.
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]system-svview
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]system-view
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]exit
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 31: Konfigurasi Link 1 pada Huawei Router

Link kedua antara Cisco dan Huawei menggunakan jaringan 10.10.11.0/30. Cisco Router meng-

gunakan interface Gi0/0 dengan IP 10.10.11.1/30, sedangkan Huawei Router menggunakan interface Ethernet1/0/4 dengan IP 10.10.11.2/30.

Tabel 3: Pengalamatan link Cisco ke Huawei (Link 2)

Perangkat	Interface	IP Address
Cisco Router	Gi0/0	10.10.11.1/30
Huawei Router	Ethernet1/0/4	10.10.11.2/30

```

Terminal
Switch x Mikrotik x VIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Router(config-if)#exhit
^
% Invalid input detected at '^' marker.

Router(config-if)#exhit
*Jun 13 05:08:46.421: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 13 05:08:47.421: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3,
changed state no shutdown
Router#
*Jun 13 05:08:49.543: %SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router#
*Jun 13 05:11:53.325: %SYS-5-CONFIG_I: Configured from console by consoleconfig
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
  
```

Gambar 32: Konfigurasi Link 2 pada Cisco Router

```

Terminal
Switch x Mikrotik x VIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[-HUAWEI]system-svview
^
Error: Unrecognized command found at '^' position.
[-HUAWEI]system-view
^
Error: Unrecognized command found at '^' position.
[-HUAWEI]exit
^
Error: Unrecognized command found at '^' position.
[-HUAWEI]interface Ethernet1/0/0
[-HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[-HUAWEI]interface Ethernet1/0/04
^
Error: Wrong parameter found at '^' position.
[-HUAWEI]interface Ethernet1/0/4
[-HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4]ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4]undo shutdown
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]commit
[-HUAWEI]

```

Gambar 33: Konfigurasi Link 2 pada Huawei Router

Link antara Huawei dan MikroTik menggunakan jaringan 10.10.20.0/30. Huawei Router menggunakan interface Ethernet1/0/3 dengan IP 10.10.20.1/30, sedangkan MikroTik menggunakan interface ether1 dengan IP 10.10.20.2/30.

Tabel 4: Pengalamatan link Huawei ke MikroTik

Perangkat	Interface	IP Address
Huawei Router	Ethernet1/0/3	10.10.20.1/30
MikroTik	ether1	10.10.20.2/30

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x

Error: Unrecognized command found at '^' position.
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]interface Ethernet1/0/04
Error: Wrong parameter found at '^' position.
[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4]ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4]undo shutdown
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]commit
[~HUAWEI]interface Ethernet1/0/43
Error: Wrong parameter found at '^' position.
[~HUAWEI]interface Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3]description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ip address 10.10.20.1 255.255.255.252
[*HUAWEI-Ethernet1/0/3]undo shutdown
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 34: Konfigurasi link Huawei ke MikroTik pada Huawei Router

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x

Jun/13/2026 04:33:18 system,error,critical login failure for user 5024241049 via local
Jun/13/2026 04:33:32 system,error,critical login failure for user admin via local

Change your password

[admin@MikroTik] > 
Expected command name (line 1 column 1)
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > ls
Bad command name ls (line 1 column 1)
[admin@MikroTik] > ls- la
Bad command name ls- (line 1 column 1)
[admin@MikroTik] > enable
Bad command name enable (line 1 column 1)
[admin@MikroTik] > /ip address address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
Syntax error (line 1 column 20)
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
[admin@MikroTik] >
```

Gambar 35: Konfigurasi link Huawei ke MikroTik pada MikroTik

Link antara Cisco dan MikroTik menggunakan jaringan 10.10.30.0/30. Cisco Router menggunakan interface Gi0/1 dengan IP 10.10.30.1/30, sedangkan MikroTik menggunakan interface ether2 dengan IP 10.10.30.2/30.

Tabel 5: Pengalamatan link Cisco ke MikroTik

Perangkat	Interface	IP Address
Cisco Router	Gi0/1	10.10.30.1/30
MikroTik	ether2	10.10.30.2/30

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router#
*Jun 13 05:11:53.325: %SYS-5-CONFIG_I: Configured from console by consoleconfig
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 13 05:12:41.299: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 13 05:12:42.300: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, c
changed state to up
Router(config)#interface gi0/1
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 13 05:17:11.816: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 13 05:17:12.816: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, c
hanged state to up

```

Gambar 36: Konfigurasi link Cisco ke MikroTik pada Cisco Router

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Jun/13/2026 04:33:18 system,error,critical login failure for user 5024241049 via
local
Jun/13/2026 04:33:32 system,error,critical login failure for user admin via local

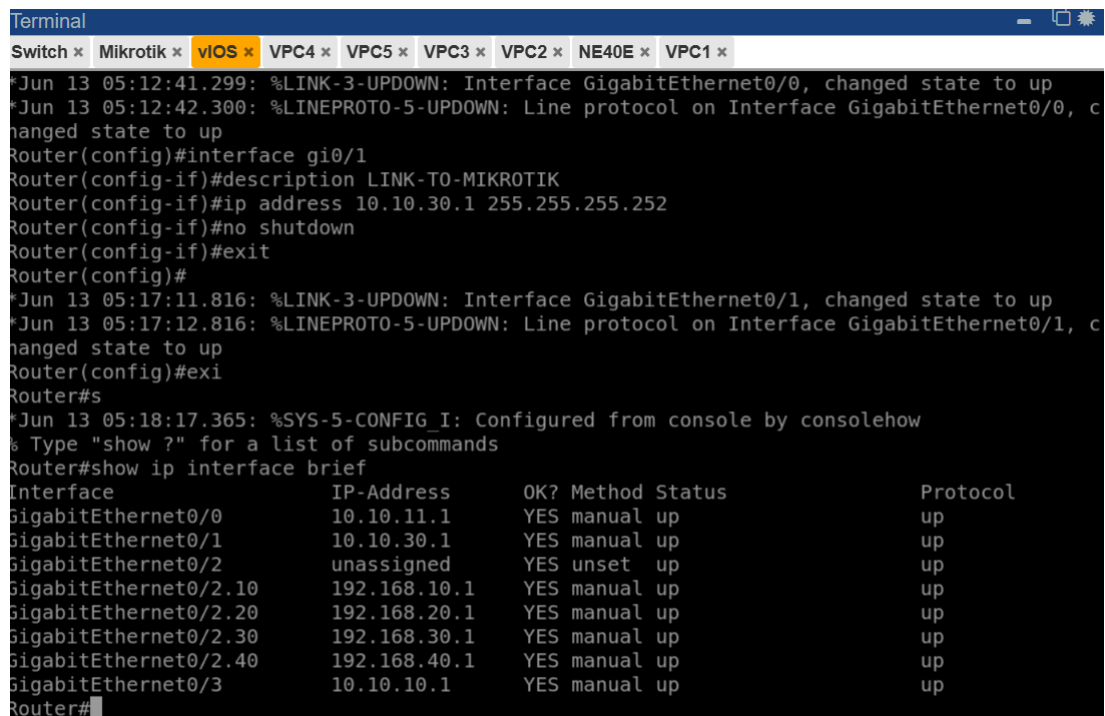
Change your password
[admin@MikroTik] > 
expected command name (line 1 column 1)
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > ls
bad command name ls (line 1 column 1)
[admin@MikroTik] > ls- la
bad command name ls- (line 1 column 1)
[admin@MikroTik] > enable
bad command name enable (line 1 column 1)
[admin@MikroTik] > /ip address address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
syntax error (line 1 column 20)
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO
[admin@MikroTik] > 

```

Gambar 37: Konfigurasi link Cisco ke MikroTik pada MikroTik

Verifikasi seluruh interface dilakukan dengan menjalankan perintah `show ip interface brief`

pada Cisco Router untuk memastikan semua interface aktif dengan IP address yang benar.



```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
*Jun 13 05:12:41.299: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 13 05:12:42.300: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, c
changed state to up
Router(config)#interface gi0/1
Router(config-if)#description LINK-TO-MIKROTIK
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 13 05:17:11.816: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 13 05:17:12.816: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, c
changed state to up
Router(config)#exi
Router#s
*Jun 13 05:18:17.365: %SYS-5-CONFIG I: Configured from console by consolehow
% Type "show ?" for a list of subcommands
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.10.11.1      YES manual up          up
GigabitEthernet0/1       10.10.30.1      YES manual up          up
GigabitEthernet0/2       unassigned      YES unset up          up
GigabitEthernet0/2.10    192.168.10.1    YES manual up          up
GigabitEthernet0/2.20    192.168.20.1    YES manual up          up
GigabitEthernet0/2.30    192.168.30.1    YES manual up          up
GigabitEthernet0/2.40    192.168.40.1    YES manual up          up
GigabitEthernet0/3       10.10.10.1      YES manual up          up
Router#
```

Gambar 38: Output show ip interface brief pada Cisco Router setelah konfigurasi link

Pada Huawei Router, dijalankan perintah display ip interface brief untuk memastikan selu-
ruh interface dalam kondisi aktif.

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
[*HUAWEI]display ip interface brief
*down: administratively down
*down: FIB overload down
*down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                IP Address/Mask      Physical  Protocol  VPN
Ethernet1/0/0            10.10.10.2/30        up        up        --
Ethernet1/0/0.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/1            unassigned           up        down      --
Ethernet1/0/1.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/2            192.168.50.1/24     up        up        --
Ethernet1/0/2.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/3            10.10.20.1/30        up        up        --
Ethernet1/0/3.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/4            10.10.11.2/30        up        up        --
Ethernet1/0/4.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/5            unassigned           up        down      --

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Interface                IP Address/Mask      Physical  Protocol  VPN
Ethernet1/0/0            10.10.10.2/30        up        up        --
Ethernet1/0/0.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/1            unassigned           up        down      --
Ethernet1/0/1.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/2            192.168.50.1/24     up        up        --
Ethernet1/0/2.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/3            10.10.20.1/30        up        up        --
Ethernet1/0/3.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/4            10.10.11.2/30        up        up        --
Ethernet1/0/4.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/5            unassigned           up        down      --
Ethernet1/0/5.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/6            unassigned           up        down      --
Ethernet1/0/6.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/7            unassigned           up        down      --
Ethernet1/0/7.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/8            unassigned           up        down      --
Ethernet1/0/8.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/9            unassigned           up        down      --
Ethernet1/0/9.4094       128.1.138.137/16    up        up        l3vpn
GigabitEthernet0/0/0     unassigned           up        down      --
LoopBack2147483647       128.1.138.137/16    up        up(s)     l3vpn
NULL0                    unassigned           up        up(s)     --

[*HUAWEI]
[*HUAWEI]

```

Gambar 39: Output display ip interface brief pada Huawei Router

Pada MikroTik, dijalankan perintah /ip address print untuk memverifikasi IP address yang telah dikonfigurasi.

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
[admin@MikroTik] > 
Expected command name (line 1 column 1)
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > 
[admin@MikroTik] > ls
Bad command name ls (line 1 column 1)
[admin@MikroTik] > ls- la
Bad command name ls- (line 1 column 1)
[admin@MikroTik] > enable
Bad command name enable (line 1 column 1)
[admin@MikroTik] > /ip address address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
Syntax error (line 1 column 20)
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] > 

```

Gambar 40: Output /ip address print pada MikroTik

Verifikasi konektivitas langsung antarrouter kemudian dilakukan. Dari Cisco Router, dilakukan ping ke setiap router tetangga langsung. Dari Huawei Router, dilakukan ping ke Cisco Router melalui dua link dan ke MikroTik. Dari MikroTik, dilakukan ping ke Cisco Router dan Huawei Router.

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
% Type "show ?" for a list of subcommands
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up up
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual up up
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/3 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#

```

Gambar 41: Ping dari Cisco Router ke Huawei Router dan MikroTik

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x

[~HUAWEI]
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=1 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=1 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=1 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=1 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
```

Gambar 42: Ping dari Huawei Router ke Cisco Router melalui Link 1 dan Link 2

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
  3 packet(s) transmitted
  3 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]
```

Gambar 43: Ping dari Huawei Router ke MikroTik


```

Terminal
Switch x Mikrotik x VIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;;; T0-HUAWEI 10.10.20.2/30 10.10.20.0 ether1
1 ;;; T0-CISCO 10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 3ms
1 10.10.20.1 56 255 1ms
2 10.10.20.1 56 255 1ms
3 10.10.20.1 56 255 1ms
4 10.10.20.1 56 255 1ms
5 10.10.20.1 56 255 3ms
6 10.10.20.1 56 255 1ms
7 10.10.20.1 56 255 1ms
sent=8 received=8 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=3ms
[admin@MikroTik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 2ms
2 10.10.30.1 56 255 1ms
sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
[admin@MikroTik] > ping 10.10.30.1

```

Gambar 44: Ping dari MikroTik ke Huawei Router dan Cisco Router

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

Praktikum keempat mencakup konfigurasi OSPF pada ketiga router dari vendor yang berbeda, yaitu Cisco, Huawei, dan MikroTik. Dua link langsung Cisco ke Huawei ditetapkan sebagai jalur utama dengan nilai *cost* 10, sedangkan jalur yang melewati MikroTik ditetapkan sebagai jalur cadangan dengan nilai *cost* 100.

Tabel 6: Nilai *cost* OSPF pada setiap link

Link	Fungsi	Cost
Cisco ke Huawei (Link 1)	Jalur utama	10
Cisco ke Huawei (Link 2)	Jalur utama	10
Cisco ke MikroTik	Jalur cadangan	100
Huawei ke MikroTik	Jalur cadangan	100

Pada Cisco Router, proses OSPF diaktifkan dengan *router-id* 1.1.1.1. Seluruh jaringan yang terhubung ke Cisco Router diumumkan ke dalam area 0, mencakup jaringan link antarrouter maupun jaringan VLAN. Nilai *cost* OSPF kemudian ditetapkan pada setiap interface: interface menuju Huawei (Link 1 dan Link 2) diberi nilai *cost* 10, sedangkan interface menuju MikroTik diberi nilai *cost* 100. Verifikasi dilakukan dengan menjalankan perintah `show ip protocols` untuk memastikan seluruh jaringan telah diumumkan. Apabila jaringan VLAN tidak muncul, konfigurasi disimpan terlebih dahulu menggunakan `wr`, kemudian dilakukan `reload` dan pengecekan ulang.

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf1
^
% Invalid input detected at '^' marker.

Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.11.0 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
```

Gambar 45: Konfigurasi OSPF pada Cisco Router

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Router(config)#router ospf1
^
% Invalid input detected at '^' marker.

Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.11.0 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#interface gi0/3
Router(config-if)#description LINK1-T0-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/1
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip ospf cost 100
Router(config-if)#exit
Router(config)#
```

Gambar 46: Penetapan nilai cost OSPF pada setiap interface Cisco Router

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Router#show ip ospf neighbor
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway          Distance      Last Update
  Distance: (default is 4)

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Incoming update filter list for all interfaces is not set
Maximum path: 32
Routing for Networks:
Routing Information Sources:
  Gateway          Distance      Last Update
Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway          Distance      Last Update
  Distance: (default is 110)

Router#
Router#

```

Gambar 47: Output show ip protocols pada Cisco Router

Pada Huawei Router, OSPF diaktifkan dengan *router-id* 2.2.2.2. Seluruh jaringan yang terhubung diumumkan ke dalam area 0. Nilai *cost* ditetapkan pada setiap interface: interface menuju Cisco (Link 1 dan Link 2) diberi nilai *cost* 10, sedangkan interface menuju MikroTik diberi nilai *cost* 100. Verifikasi dilakukan dengan perintah `display ospf peer` untuk memastikan Huawei membentuk *peer* dengan Cisco melalui dua link utama. Tabel routing OSPF yang diterima dari Cisco juga diperiksa menggunakan perintah `display ip routing-table protocol ospf` dan `display ospf routing`.

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ospf1 router-id 2.2.2.2
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1]area0,
^
Error: Unrecognized command found at '^' position.
[*HUAWEI-ospf-1]area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0]quit
[*HUAWEI-ospf-1]exit
^
Error: Unrecognized command found at '^' position.
[*HUAWEI-ospf-1]quit
[*HUAWEI]
```

Gambar 48: Konfigurasi OSPF pada Huawei Router

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
^
Error: Unrecognized command found at '^' position.
[*HUAWEI-ospf-1]area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0]quit
[*HUAWEI-ospf-1]exit
^
Error: Unrecognized command found at '^' position.
[*HUAWEI-ospf-1]quit
[*HUAWEI]display ospf peer
[*HUAWEI]interface Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ospf cost 10
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4]ospf cost 10
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3]description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ospf cost 100
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 49: Penetapan nilai cost OSPF pada setiap interface Huawei Router

```
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
[*HUAWEI]commit
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1          Address: 10.10.10.1
  State: Full              Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1          BDR: 10.10.10.2        MTU: 1500
  Dead timer due in 31 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m23s
  Neighbor Up Time : 2026-06-13 05:33:24
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1          Address: 10.10.11.1
  State: Full              Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1          BDR: 10.10.11.2        MTU: 1500
  Dead timer due in 39 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m23s
  Neighbor Up Time : 2026-06-13 05:33:24
```

Gambar 50: Output display ospf peer pada Huawei Router

```

Terminal
Switch * Mikrotik * vIOS * VPC4 * VPC5 * VPC3 * VPC2 * NE40E * VPC1 *
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface
10.10.30.0/30       OSPF   10   110        D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   110        D   10.10.10.1    Ethernet1/0/0
192.168.10.0/24     OSPF   10   11         D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   11         D   10.10.10.1    Ethernet1/0/0
192.168.20.0/24     OSPF   10   11         D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   11         D   10.10.10.1    Ethernet1/0/0
192.168.30.0/24     OSPF   10   11         D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   11         D   10.10.10.1    Ethernet1/0/0
192.168.40.0/24     OSPF   10   11         D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   11         D   10.10.10.1    Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface

Terminal
Switch * Mikrotik * vIOS * VPC4 * VPC5 * VPC3 * VPC2 * NE40E * VPC1 *
OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface
10.10.30.0/30       OSPF   10   110        D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   110        D   10.10.10.1    Ethernet1/0/0
192.168.10.0/24     OSPF   10   11         D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   11         D   10.10.10.1    Ethernet1/0/0
192.168.20.0/24     OSPF   10   11         D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   11         D   10.10.10.1    Ethernet1/0/0
192.168.30.0/24     OSPF   10   11         D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   11         D   10.10.10.1    Ethernet1/0/0
192.168.40.0/24     OSPF   10   11         D   10.10.11.1    Ethernet1/0/4
                   OSPF   10   11         D   10.10.10.1    Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface
10.10.10.0/30       OSPF   10   10         10.10.10.2    Ethernet1/0/0
10.10.11.0/30       OSPF   10   10         10.10.11.2    Ethernet1/0/4
10.10.20.0/30       OSPF   10  100        10.10.20.1    Ethernet1/0/3
192.168.50.0/24     OSPF   10   1          192.168.50.1  Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]

```

Gambar 51: Tabel routing OSPF yang diterima Huawei Router dari Cisco

```

Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
[~HUAWEI]
[~HUAWEI]display ospf routing

      OSPF Process 1 with Router ID 2.2.2.2
      Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9  Inter Area: 0  ASE: 0  NSSA: 0

      OSPF Process 65534 with Router ID 128.1.138.137
Terminal
Switch x Mikrotik x vIOS x VPC4 x VPC5 x VPC3 x VPC2 x NE40E x VPC1 x
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9  Inter Area: 0  ASE: 0  NSSA: 0

      OSPF Process 65534 with Router ID 128.1.138.137
      Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
128.1.138.137/32  0         Direct    128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1  Inter Area: 0  ASE: 0  NSSA: 0
[~HUAWEI]

```

Gambar 52: Output display ospf routing pada Huawei Router

Pada MikroTik yang berperan sebagai jalur cadangan, *router-id* OSPF ditetapkan sebagai 3.3.3.3. Jaringan 10.10.20.0/30 dan 10.10.30.0/30 diumumkan ke OSPF pada area backbone. Nilai *cost* pada kedua interface MikroTik ditetapkan sebesar 100. Verifikasi dilakukan dengan perintah `/routing ospf neighbor print` untuk memastikan Cisco (1.1.1.1) dan Huawei (2.2.2.2) muncul sebagai *neighbor*. Tabel routing OSPF juga diperiksa menggunakan perintah `/ip route print where ospf` untuk memastikan MikroTik memperoleh rute menuju seluruh VLAN dan LAN Huawei.

```

Terminal
vIOS x Mikrotik x

[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=14s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=21s
[admin@MikroTik] >

```

Gambar 53: Output `/routing ospf neighbor print` pada MikroTik

```

Terminal
vIOS x Mikrotik x

[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=14s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=21s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 Ado  10.10.10.0/30          10.10.30.1    110
      10.10.20.1
1 Ado  10.10.11.0/30          10.10.30.1    110
      10.10.20.1
2 Ado  192.168.10.0/24        10.10.30.1    110
3 Ado  192.168.20.0/24        10.10.30.1    110
4 Ado  192.168.30.0/24        10.10.30.1    110
5 Ado  192.168.40.0/24        10.10.30.1    110
6 Ado  192.168.50.0/24        10.10.20.1    110
[admin@MikroTik] >

```

Gambar 54: Tabel routing OSPF pada MikroTik

Verifikasi OSPF *neighbor* dilakukan pada seluruh router. Pada Cisco Router, Huawei diharapkan muncul dua kali (melalui dua link utama) dan MikroTik muncul satu kali. Pada Huawei Router, Cisco diharapkan muncul dua kali dan MikroTik muncul satu kali. Pada MikroTik, Cisco dan Huawei keduanya diharapkan muncul sebagai *neighbor*.


```

Terminal
vIOS x Mikrotik x

Router#
*Jun 13 05:39:50.464: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOA
DING to FULL, Loading Done
*Jun 13 05:40:33.501: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOA
DING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address       Interface
3.3.3.3          1     FULL/BDR        00:00:32    10.10.30.2    GigabitEthernet0/1
2.2.2.2          1     FULL/BDR        00:00:31    10.10.11.2    GigabitEthernet0/0
2.2.2.2          1     FULL/BDR        00:00:33    10.10.10.2    GigabitEthernet0/3
Router#

```

Gambar 55: Output show ip ospf neighbor pada Cisco Router

```

Terminal
vIOS x Mikrotik x NE40E x VPC1 x
(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full           Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1        BDR: 10.10.10.2      MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5
  Neighbor is up for 00h09m15s
  Neighbor Up Time : 2026-06-13 05:33:24
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
  State: Full           Mode:Nbr is Master    Priority: 1
  DR: 10.10.20.1        BDR: 10.10.20.2      MTU: 1500
  Dead timer due in 32 sec
  Retrans timer interval: 5
  Neighbor is up for 00h01m34s
  Neighbor Up Time : 2026-06-13 05:41:05
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full           Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1        BDR: 10.10.11.2      MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5

```

Gambar 56: Output display ospf peer pada Huawei Router setelah MikroTik terkonfigurasi

```

Terminal
VIOS x Mikrotik x NE40E x VPC1 x
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=14s

1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=21s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 ADo  10.10.10.0/30      10.10.30.1    110
      10.10.20.1
1 ADo  10.10.11.0/30      10.10.30.1    110
      10.10.20.1
2 ADo  192.168.10.0/24      10.10.30.1    110
3 ADo  192.168.20.0/24      10.10.30.1    110
4 ADo  192.168.30.0/24      10.10.30.1    110
5 ADo  192.168.40.0/24      10.10.30.1    110
6 ADo  192.168.50.0/24      10.10.20.1    110
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m27s

1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m34s
[admin@MikroTik] >

```

Gambar 57: Output `/routing ospf neighbor print` pada MikroTik

Verifikasi tabel routing OSPF dilakukan pada seluruh router. Pada Cisco Router, rute menuju LAN Huawei diperiksa menggunakan perintah `show ip route 192.168.50.0`. Kemunculan dua *next-hop* menandakan OSPF memanfaatkan kedua link utama secara bersamaan. Seluruh rute OSPF pada Cisco Router kemudian diperiksa menggunakan `show ip route ospf`.

```

Terminal
vIOS x Mikrotik x NE40E x VPC1 x

Router#
*Jun 13 05:39:50.464: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOA
DING to FULL, Loading Done
*Jun 13 05:40:33.501: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOA
DING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1     FULL/BDR        00:00:32    10.10.30.2     GigabitEthernet0/1
2.2.2.2          1     FULL/BDR        00:00:31    10.10.11.2     GigabitEthernet0/0
2.2.2.2          1     FULL/BDR        00:00:33    10.10.10.2     GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:10:31 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:10:31 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:10:31 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#

```

Gambar 58: Rute Cisco Router menuju LAN Huawei 192.168.50.0/24

```

Terminal
vIOS x Mikrotik x NE40E x VPC1 x

Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1     FULL/BDR        00:00:32    10.10.30.2     GigabitEthernet0/1
2.2.2.2          1     FULL/BDR        00:00:31    10.10.11.2     GigabitEthernet0/0
2.2.2.2          1     FULL/BDR        00:00:33    10.10.10.2     GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:10:31 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:10:31 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:10:31 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O        10.10.20.0/30 [110/110] via 10.10.11.2, 00:10:45, GigabitEthernet0/0
          [110/110] via 10.10.10.2, 00:10:45, GigabitEthernet0/3
O        192.168.50.0/24 [110/11] via 10.10.11.2, 00:10:45, GigabitEthernet0/0
          [110/11] via 10.10.10.2, 00:10:45, GigabitEthernet0/3
Router#

```

Gambar 59: Seluruh rute OSPF pada Cisco Router

```

Terminal
vIOS x Mikrotik x NE40E x VPC1 x
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
_public_ Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface
10.10.30.0/30       OSPF   10   110       D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   110       D    10.10.10.1     Ethernet1/0/0
192.168.10.0/24      OSPF   10   11         D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   11         D    10.10.10.1     Ethernet1/0/0
192.168.20.0/24      OSPF   10   11         D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   11         D    10.10.10.1     Ethernet1/0/0
192.168.30.0/24      OSPF   10   11         D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   11         D    10.10.10.1     Ethernet1/0/0
192.168.40.0/24      OSPF   10   11         D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   11         D    10.10.10.1     Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface
10.10.10.0/30       OSPF   10   10         10.10.10.2     Ethernet1/0/0
10.10.11.0/30       OSPF   10   10         10.10.11.2     Ethernet1/0/4
10.10.20.0/30       OSPF   10  100        10.10.20.1     Ethernet1/0/3
192.168.50.0/24     OSPF   10   1          192.168.50.1   Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]

```

Gambar 60: Seluruh rute OSPF pada Huawei Router

```

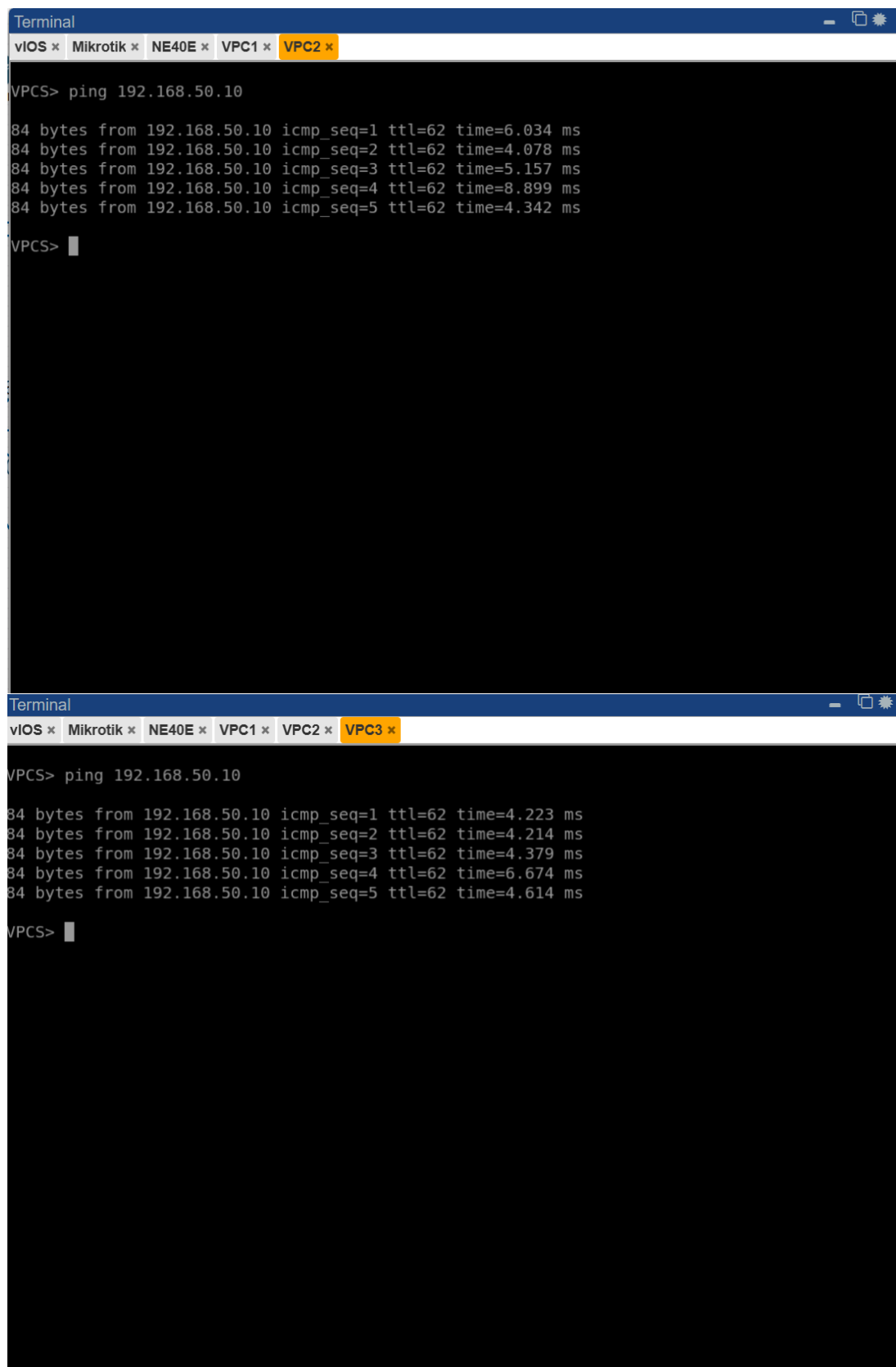
Terminal
vIOS x Mikrotik x NE40E x VPC1 x
0 ADo 10.10.10.0/30      10.10.30.1      110
      10.10.20.1
1 ADo 10.10.11.0/30     10.10.30.1      110
      10.10.20.1
2 ADo 192.168.10.0/24   10.10.30.1      110
3 ADo 192.168.20.0/24   10.10.30.1      110
4 ADo 192.168.30.0/24   10.10.30.1      110
5 ADo 192.168.40.0/24   10.10.30.1      110
6 ADo 192.168.50.0/24   10.10.20.1      110
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m27s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m34s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#    DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30      10.10.30.1  110
      10.10.20.1
1 ADo 10.10.11.0/30     10.10.30.1  110
      10.10.20.1
2 ADo 192.168.10.0/24   10.10.30.1  110
3 ADo 192.168.20.0/24   10.10.30.1  110
4 ADo 192.168.30.0/24   10.10.30.1  110
5 ADo 192.168.40.0/24   10.10.30.1  110
6 ADo 192.168.50.0/24   10.10.20.1  110
[admin@MikroTik] >

```

Gambar 61: Seluruh rute OSPF pada MikroTik

Pengujian konektivitas *end-to-end* dilakukan dengan melakukan ping dari setiap PC VLAN ke PC

LAN Huawei (192.168.50.10) untuk memastikan OSPF berhasil mendistribusikan rute ke seluruh jaringan.



The image displays two screenshots of a terminal window. The top screenshot shows the terminal with tabs for vIOS, Mikrotik, NE40E, VPC1, and VPC2. The command 'ping 192.168.50.10' has been executed, resulting in five successful ping responses with varying times. The bottom screenshot shows the same terminal with an additional tab for VPC3. The command 'ping 192.168.50.10' is again executed, showing five successful ping responses with different times.

```
Terminal
vIOS x Mikrotik x NE40E x VPC1 x VPC2 x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.034 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.078 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.157 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.899 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.342 ms
VPCS>

Terminal
vIOS x Mikrotik x NE40E x VPC1 x VPC2 x VPC3 x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.223 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.214 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.379 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.674 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.614 ms
VPCS>
```

The image consists of two screenshots of a terminal window. The top screenshot shows the terminal with tabs for vIOS, Mikrotik, NE40E, VPC1, VPC2, VPC3, and VPC4 (highlighted). The command 'VPCS> ping 192.168.50.10' has been executed, resulting in five successful ping responses from 192.168.50.10 with varying times (4.647 ms to 8.664 ms). The bottom screenshot shows the same terminal with tabs for vIOS, Mikrotik, NE40E, VPC1, VPC2, VPC3, VPC4, and VPC5 (highlighted). The command 'VPCS> ping 192.168.50.10' has been executed, resulting in five successful ping responses from 192.168.50.10 with varying times (13.117 ms to 3.891 ms).

```
terminal
vIOS x Mikrotik x NE40E x VPC1 x VPC2 x VPC3 x VPC4 x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.647 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.872 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.643 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.808 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.664 ms
VPCS>

Terminal
vIOS x Mikrotik x NE40E x VPC1 x VPC2 x VPC3 x VPC4 x VPC5 x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=13.117 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.494 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.555 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=4.240 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.891 ms
VPCS>
```

Gambar 62: Ping dari PC masing-masing VLAN ke PC LAN Huawei 192.168.50.10

Dari PC LAN Huawei, dilakukan pula ping ke PC di setiap VLAN Cisco untuk memverifikasi konektivitas dua arah.

```
Terminal
vIOS x Mikrotik x NE40E x VPC1 x VPC2 x VPC3 x VPC4 x VPC5 x

VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=3.893 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=4.125 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=5.017 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=6.617 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=3.133 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=4.306 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=4.531 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=3.233 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=5.430 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=4.104 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=4.751 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=4.872 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.345 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=4.243 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=3.988 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=3.410 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=3.746 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=4.884 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=7.829 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=4.574 ms
```

Gambar 63: Ping dari PC LAN Huawei ke PC di setiap VLAN Cisco

1.5 Praktikum 5: Pengujian Failover OSPF

Praktikum kelima menguji kemampuan OSPF dalam berpindah jalur secara otomatis ketika link utama mengalami gangguan. Pengujian dilaksanakan dalam empat skenario berurutan.

Pada skenario pertama, kondisi jaringan dalam keadaan normal dengan kedua link Cisco ke Huawei aktif. Rute dari Cisco Router menuju LAN Huawei diperiksa menggunakan perintah `show ip route 192.168.50.0`. Kemudian dilakukan `traceroute` dari PC VLAN ke PC LAN Huawei. Trafik diharapkan melewati jalur langsung Cisco ke Huawei.

```
vIOS x Mikrotik x NE40E x VPC1 x VPC2 x VPC3 x VPC4 x VPC5 x

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.034 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.078 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.157 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.899 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.342 ms

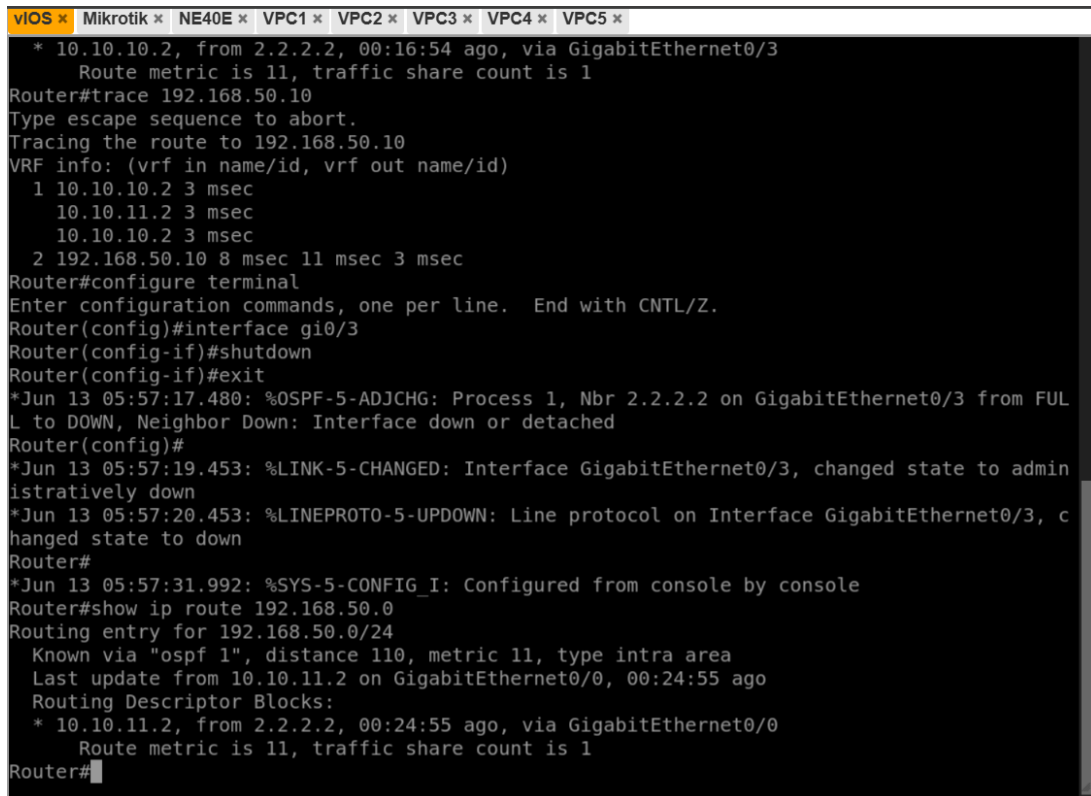
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.158 ms 2.161 ms 3.070 ms
 2 10.10.11.2 5.301 ms 3.436 ms 3.160 ms
 3 *192.168.50.10 7.035 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> █
```

Gambar 64: Traceroute pada kondisi normal, dua link Cisco ke Huawei aktif

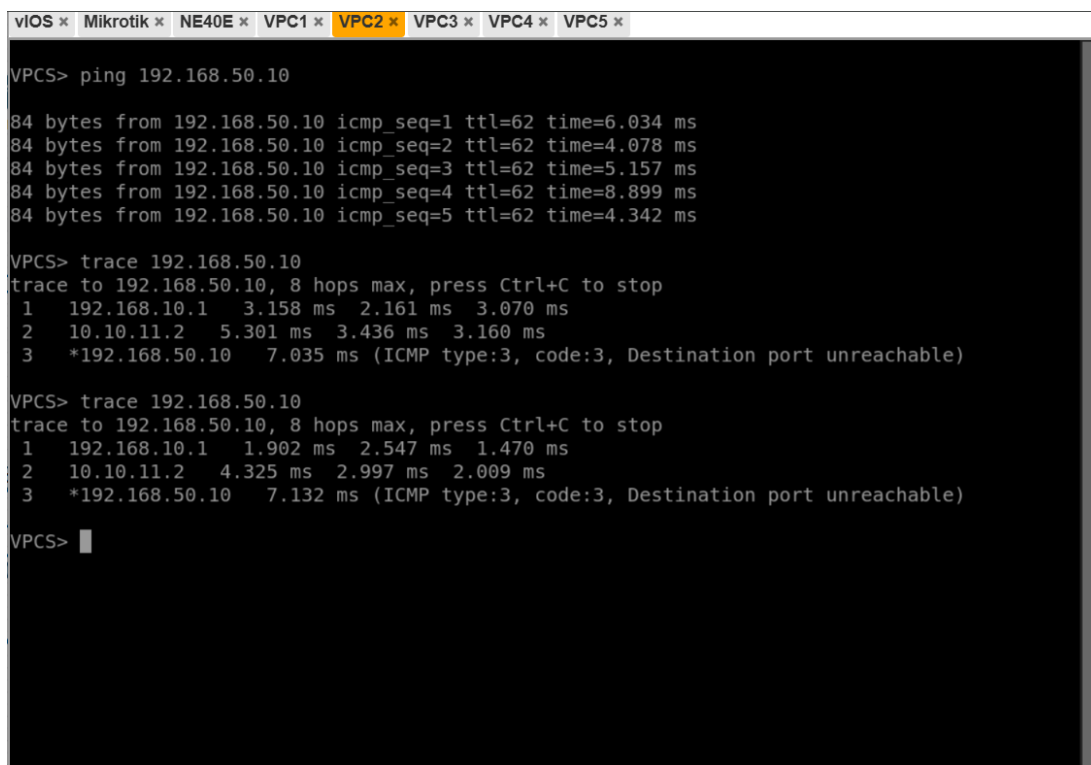
Pada skenario kedua, interface `gi0/3` pada Cisco Router dinonaktifkan untuk mensimulasikan kegagalan Link 1. Setelah itu, rute kembali diperiksa dan `traceroute` diulang untuk memastikan trafik

masih melewati jalur langsung melalui Link 2 yang tetap aktif.



```
vIOS * Mikrotik * NE40E * VPC1 * VPC2 * VPC3 * VPC4 * VPC5 *
* 10.10.10.2, from 2.2.2.2, 00:16:54 ago, via GigabitEthernet0/3
  Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.10.2 3 msec
    10.10.11.2 3 msec
    10.10.10.2 3 msec
  2 192.168.50.10 8 msec 11 msec 3 msec
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#shutdown
Router(config-if)#exit
*Jun 13 05:57:17.480: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Interface down or detached
Router(config)#
*Jun 13 05:57:19.453: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun 13 05:57:20.453: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
Router#
*Jun 13 05:57:31.992: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:24:55 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:24:55 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#
```

Gambar 65: Pemeriksaan rute dari Cisco Router menuju LAN Huawei setelah Link 1 dinonaktifkan



```
vIOS * Mikrotik * NE40E * VPC1 * VPC2 * VPC3 * VPC4 * VPC5 *
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.034 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.078 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.157 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.899 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.342 ms

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.158 ms 2.161 ms 3.070 ms
 2 10.10.11.2 5.301 ms 3.436 ms 3.160 ms
 3 *192.168.50.10 7.035 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 1.902 ms 2.547 ms 1.470 ms
 2 10.10.11.2 4.325 ms 2.997 ms 2.009 ms
 3 *192.168.50.10 7.132 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 66: Traceroute saat satu link Cisco ke Huawei dinonaktifkan

Pada skenario ketiga, interface gi0/0 pada Cisco Router juga dinonaktifkan sehingga kedua link utama tidak aktif. Rute kembali diperiksa dan *traceroute* diulang. Rute diharapkan beralih ke jalur

cadangan melalui MikroTik dengan urutan hop: PC VLAN, Cisco Router, MikroTik, Huawei Router, dan LAN Huawei.

```
viOS x Mikrotik x NE40E x VPC1 x VPC2 x VPC3 x VPC4 x VPC5 x
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#shutdown
Router(config-if)#interface gi0/0
*Jun 13 05:59:11.572: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
*Jun 13 05:59:13.544: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
*Jun 13 05:59:14.545: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
Router#
*Jun 13 05:59:17.725: %SYS-5-CONFIG_I: Configured from console by console^Z
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#show ip route 192.168.50.0
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#show
*Jun 13 06:00:05.351: %SYS-5-CONFIG_I: Configured from console by console^Z
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:01:02 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:01:02 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#
```

Gambar 67: Pemeriksaan rute dari Cisco Router menuju LAN Huawei setelah kedua link utama dinonaktifkan

```
viOS x Mikrotik x NE40E x VPC1 x VPC2 x VPC3 x VPC4 x VPC5 x
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.034 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.078 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.157 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.899 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.342 ms

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.158 ms  2.161 ms  3.070 ms
 2  10.10.11.2     5.301 ms  3.436 ms  3.160 ms
 3  *192.168.50.10  7.035 ms  (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    1.902 ms  2.547 ms  1.470 ms
 2  10.10.11.2     4.325 ms  2.997 ms  2.009 ms
 3  *192.168.50.10  7.132 ms  (ICMP type:3, code:3, Destination port unreachable)

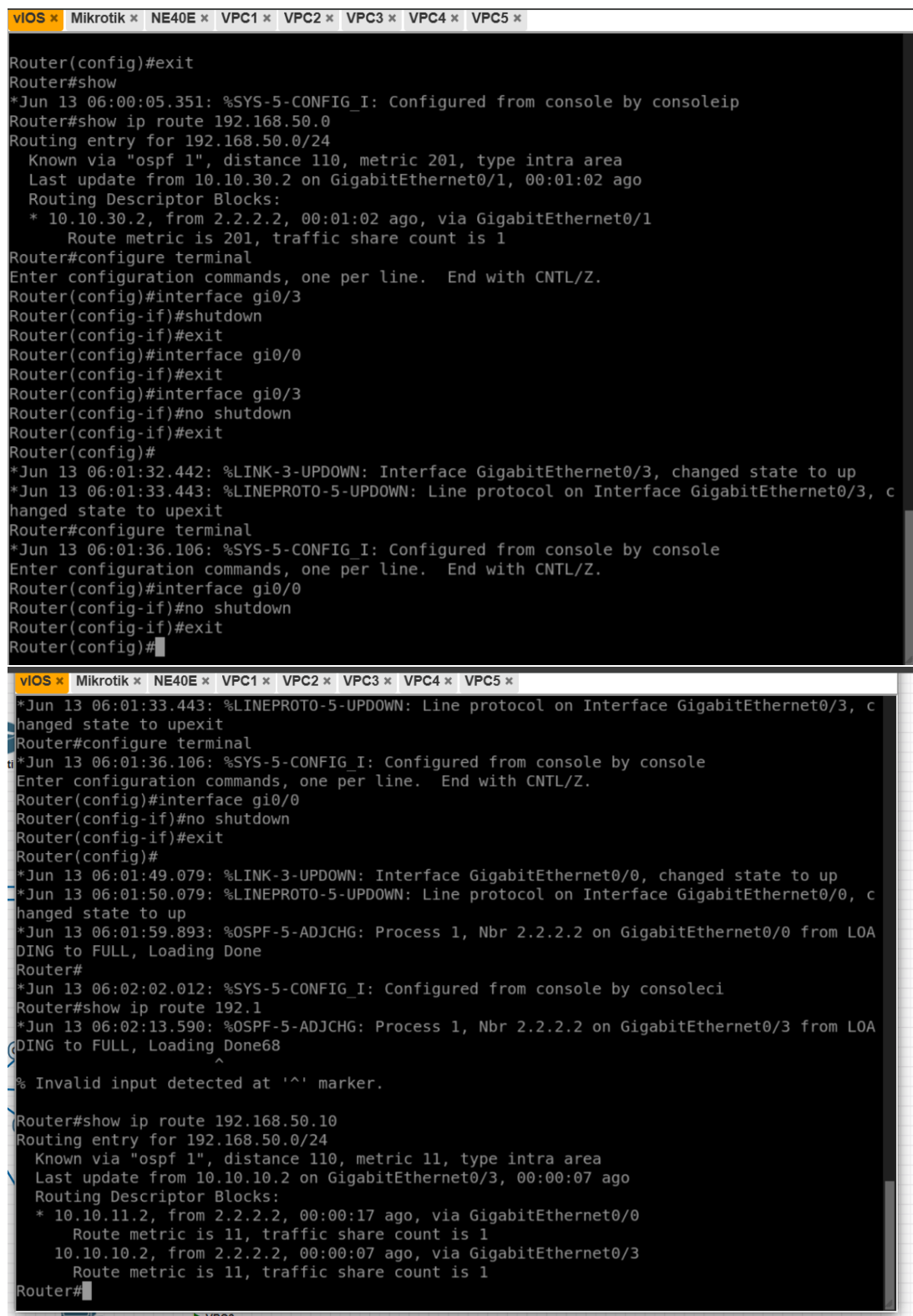
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.871 ms  2.631 ms  1.895 ms
 2  10.10.30.2     2.988 ms  1.716 ms  1.524 ms
 3  10.10.20.1     5.472 ms  3.061 ms  3.213 ms
 4  *192.168.50.10  3.721 ms  (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 68: Traceroute saat kedua link Cisco ke Huawei dinonaktifkan, jalur beralih melalui MikroTik

Pada skenario keempat, interface gi0/3 dan gi0/0 diaktifkan kembali pada Cisco Router. Ru-

te kembali diperiksa dan *traceroute* diulang untuk memastikan trafik kembali melewati jalur utama langsung Cisco ke Huawei, bukan melalui MikroTik.



```
Router(config)#exit
Router#show
*Jun 13 06:00:05.351: %SYS-5-CONFIG_I: Configured from console by consoleip
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:01:02 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:01:02 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#interface gi0/0
Router(config-if)#exit
Router(config)#interface gi0/3
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 13 06:01:32.442: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 13 06:01:33.443: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, c
changed state to upexit
Router#configure terminal
*Jun 13 06:01:36.106: %SYS-5-CONFIG_I: Configured from console by console
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#

*Jun 13 06:01:33.443: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, c
changed state to upexit
Router#configure terminal
*Jun 13 06:01:36.106: %SYS-5-CONFIG_I: Configured from console by console
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 13 06:01:49.079: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 13 06:01:50.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, c
changed state to up
*Jun 13 06:01:59.893: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOA
DING to FULL, Loading Done
Router#
*Jun 13 06:02:02.012: %SYS-5-CONFIG_I: Configured from console by consoleci
Router#show ip route 192.1
*Jun 13 06:02:13.590: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOA
DING to FULL, Loading Done68
^
% Invalid input detected at '^' marker.
Router#show ip route 192.168.50.10
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.10.2 on GigabitEthernet0/3, 00:00:07 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:00:17 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    10.10.10.2, from 2.2.2.2, 00:00:07 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#
```

Gambar 69: Pemeriksaan rute dari Cisco Router menuju LAN Huawei setelah kedua link utama diaktifkan kembali

```
vIOS x Mikrotik x NE40E x VPC1 x VPC2 x VPC3 x VPC4 x VPC5 x
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.034 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.078 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.157 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.899 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.342 ms

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.158 ms  2.161 ms  3.070 ms
 2  10.10.11.2     5.301 ms  3.436 ms  3.160 ms
 3  *192.168.50.10  7.035 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    1.902 ms  2.547 ms  1.470 ms
 2  10.10.11.2     4.325 ms  2.997 ms  2.009 ms
 3  *192.168.50.10  7.132 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.871 ms  2.631 ms  1.895 ms
 2  10.10.30.2     2.988 ms  1.716 ms  1.524 ms
 3  10.10.20.1     5.472 ms  3.061 ms  3.213 ms
 4  *192.168.50.10  3.721 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.448 ms  4.022 ms  1.852 ms
 2  10.10.11.2     5.969 ms  5.097 ms  3.218 ms
 3  *192.168.50.10 13.509 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> █
```

Gambar 70: Traceroute setelah kedua link Cisco ke Huawei diaktifkan kembali

2 Analisis Hasil Percobaan

2.1 Analisis Praktikum 1: Konfigurasi Cisco Switch

Output perintah `show vlan brief` menunjukkan bahwa VLAN 10, 20, 30, dan 40 berhasil dibuat dengan status *active*, sesuai dengan konsep VLAN sebagai mekanisme pembagian satu jaringan fisik menjadi beberapa segmen logis yang terisolasi berdasarkan VLAN ID. Konfigurasi *access port* pada setiap interface yang mengarah ke PC menyebabkan trafik yang melewatinya bersifat *untagged* dan hanya dikenali sebagai anggota VLAN yang telah ditetapkan. Output perintah `show interfaces trunk` membuktikan bahwa interface Gi0/0 berhasil dikonfigurasi sebagai *trunk port* dengan enkapsulasi 802.1Q sehingga mampu membawa keempat VLAN secara bersamaan melalui satu link fisik menuju Cisco Router. Kesesuaian antara konfigurasi yang diterapkan dan hasil verifikasi menunjukkan bahwa mekanisme *tagging* VLAN pada trunk berjalan sesuai dengan teori.

2.2 Analisis Praktikum 2: Konfigurasi Cisco Router

Penerapan metode ROAS (*Router on a Stick*) melalui *subinterface* memungkinkan satu interface fisik Cisco Router menangani routing antar-VLAN tanpa memerlukan interface fisik terpisah untuk setiap VLAN. Setiap *subinterface* dikonfigurasi dengan enkapsulasi 802.1Q yang sesuai dengan VLAN ID pada trunk sehingga paket yang masuk dapat diarahkan ke VLAN yang tepat berdasarkan label tag-nya. Konfigurasi DHCP *server* berhasil mendistribusikan IP address secara otomatis kepada PC VLAN 10 dan VLAN 20, yang dibuktikan melalui keberhasilan ping ke *gateway* maupun ping antar-VLAN. Hasil ini selaras dengan teori *inter-VLAN routing*, yaitu router berperan sebagai penghubung antarsegmen jaringan logis yang sebelumnya terisolasi pada level switch.

2.3 Analisis Praktikum 3: Konfigurasi Huawei Router dan IP Antarrouter

Pengalamatan IP pada keempat link antarrouter berhasil dilakukan tanpa konflik subnet, dengan masing-masing link menggunakan subnet /30 yang sesuai untuk koneksi *point-to-point*. Keberhasilan ping antarrouter tetangga sebelum OSPF dikonfigurasi membuktikan bahwa konektivitas layer 3 pada seluruh link telah berfungsi dengan baik, yang merupakan prasyarat penting karena OSPF tidak dapat membentuk hubungan *neighbor* apabila konektivitas dasar antarperangkat belum tersedia. PC pada LAN Huawei juga berhasil melakukan ping ke *gateway*, yang menunjukkan bahwa pengalamatan pada sisi Huawei telah sesuai dengan tabel *addressing* yang direncanakan.

2.4 Analisis Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

Pembentukan *neighbor* OSPF antara Cisco Router, Huawei Router, dan MikroTik berhasil dilakukan meskipun ketiga perangkat berasal dari vendor yang berbeda. Hal ini sejalan dengan sifat OSPF sebagai protokol *open standard* yang tidak terikat pada satu vendor tertentu, sehingga dapat diimplementasikan secara *multivendor* selama parameter dasar seperti *area* dan tipe jaringan pada interface telah diselaraskan. Penetapan nilai *cost* 10 pada dua link Cisco ke Huawei dan nilai *cost* 100 pada link melalui MikroTik menghasilkan jalur utama yang diprioritaskan oleh algoritma Dijkstra karena memiliki akumulasi *cost* yang lebih rendah. Keberhasilan ping dari PC di seluruh VLAN Cisco ke PC LAN Huawei dan sebaliknya membuktikan bahwa OSPF berhasil mendistribusikan informasi routing ke seluruh jaringan lintas vendor, sesuai dengan prinsip *link-state routing*.

2.5 Analisis Praktikum 5: Pengujian Failover OSPF

Hasil *traceroute* pada kondisi normal menunjukkan bahwa trafik melewati jalur langsung Cisco ke Huawei dengan nilai *cost* terendah, sesuai dengan prinsip OSPF dalam memilih jalur berdasarkan total *cost* pada SPF *tree*. Ketika satu link utama dinonaktifkan, OSPF secara otomatis menghapus rute melalui link tersebut dan mengalihkan trafik ke link utama yang masih aktif tanpa intervensi manual. Ketika kedua link utama dinonaktifkan, trafik beralih ke jalur cadangan melalui MikroTik dengan urutan hop: PC VLAN, Cisco Router, MikroTik, Huawei Router, dan LAN Huawei. Peralihan ini terjadi karena OSPF menghitung ulang SPF *tree* setelah mendeteksi perubahan topologi melalui mekanisme *flooding* LSA. Ketika kedua link utama diaktifkan kembali, trafik secara otomatis kembali ke jalur langsung Cisco ke Huawei karena nilai *cost*-nya lebih rendah. Keseluruhan skenario ini membuktikan bahwa mekanisme *failover* OSPF berbasis nilai *cost* berjalan secara otomatis dan konsisten dengan teori *convergence* pada protokol *link-state routing*.

2.6 Faktor yang Memengaruhi Hasil Percobaan

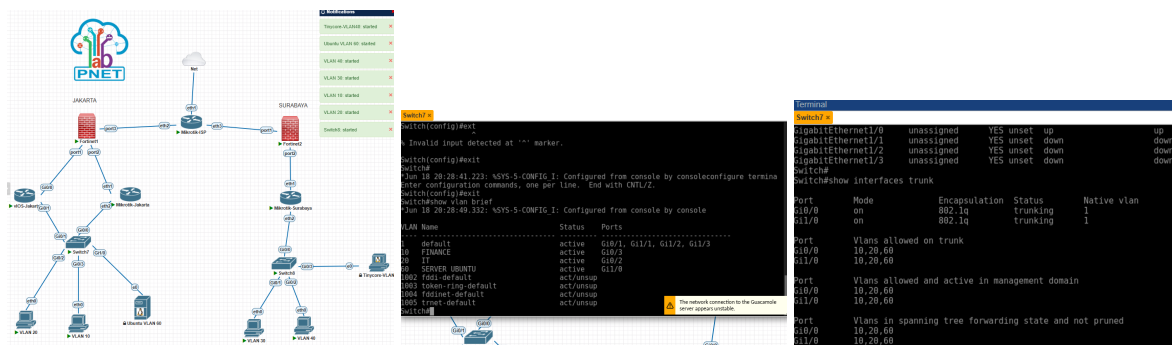
Beberapa faktor yang berpotensi memengaruhi keberhasilan percobaan antara lain kesesuaian nomor interface fisik pada simulator PNETLab dengan topologi yang direncanakan, ketelitian dalam penulisan VLAN ID dan subnet pada setiap perangkat, serta waktu *convergence* OSPF yang memerlukan beberapa detik setelah perubahan topologi sebelum tabel routing diperbarui sepenuhnya. Kesalahan kecil seperti subnet mask yang tidak konsisten pada kedua ujung link atau VLAN yang tidak diizinkan pada trunk dapat mengakibatkan kegagalan konektivitas meskipun konfigurasi routing sudah benar. Secara keseluruhan, seluruh praktikum berhasil dilaksanakan sesuai dengan tujuan

modul, yaitu membangun dan menguji jaringan *multivendor* yang terintegrasi menggunakan VLAN, *inter-VLAN routing*, dan OSPF dengan mekanisme *failover* otomatis.

3 Hasil Tugas Modul

Tugas modul berupa implementasi jaringan *enterprise* berskala lebih besar yang menghubungkan kantor pusat (HQ) di Jakarta dengan kantor cabang (*Branch*) di Surabaya menggunakan VRRP, ISC-DHCP Server, FortiGate, GRE Tunnel, dan OSPF. Tugas dibagi menjadi sepuluh bagian yang dikerjakan secara bertahap, dengan ringkasan hasil sebagai berikut.

1. **Konfigurasi Cisco Switch Jakarta.** VLAN 10, 20, dan 60 berhasil dibuat dan diaktifkan pada Cisco Switch Jakarta. Port menuju klien VLAN 10 dan VLAN 20 dikonfigurasi sebagai *access port*, sedangkan port menuju Ubuntu Server dikonfigurasi sebagai *access port* VLAN 60. Link menuju Cisco Router dan Mikrotik Router Jakarta dikonfigurasi sebagai *trunk* yang membawa ketiga VLAN tersebut. Gambar di bawah ini membuktikan topologi yang digunakan, hasil perintah `show vlan brief` yang menampilkan port access, serta `show interfaces trunk` yang mengonfirmasi status trunking.



Gambar 71: Bukti (1) Topologi jaringan, (2) Daftar VLAN & port access, dan (3) Status antarmuka trunk pada Switch Jakarta.

2. **Konfigurasi Cisco Router Jakarta.** *Subinterface* untuk VLAN 10, 20, dan 60 berhasil dibuat dengan pengalamatan IP yang sesuai. VRRP berhasil dikonfigurasi dengan Cisco Router sebagai *master* untuk VLAN 10 dan 60. DHCP *relay* berhasil diarahkan ke Ubuntu Server Jakarta, dan koneksi ke FortiGate Jakarta diverifikasi melalui ping. Gambar berikut menunjukkan ringkasan IP interface, status VRRP (Master), konfigurasi subinterface, dan keberhasilan *ping* ke FortiGate.

```

Terminal
Switch7 x vIOS-Jakarta x
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.10.100.1
Router(config)#end
Router#write
*Jun 18 21:07:08.928: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
Router#
*Jun 18 21:07:14.236: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk
Please wait...
*Jun 18 21:07:14.918: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Router#show ip interface br
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.100.2 YES manual up up
GigabitEthernet0/1 unassigned YES unset up up
GigabitEthernet0/1.10 192.168.10.2 YES manual up up
GigabitEthernet0/1.20 192.168.20.2 YES manual up up
GigabitEthernet0/1.60 192.168.60.2 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
GigabitEthernet0/3 unassigned YES unset administratively down down
Router#

Terminal
Switch7 x vIOS-Jakarta x
media-type rj45
end

Router#show running-config | section interface
interface GigabitEthernet0/0
ip address 10.10.100.2 255.255.255.252
duplex auto
speed auto
media-type rj45
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
media-type rj45
interface GigabitEthernet0/1.10
encapsulation dot1Q 10
ip address 192.168.10.2 255.255.255.0
ip helper-address 192.168.60.10
vrrp 10 ip 192.168.10.1
vrrp 10 priority 110
interface GigabitEthernet0/1.20
encapsulation dot1Q 20
ip address 192.168.20.2 255.255.255.0
ip helper-address 192.168.60.10
vrrp 20 ip 192.168.20.1
vrrp 20 priority 90
interface GigabitEthernet0/1.60
encapsulation dot1Q 60
ip address 192.168.60.2 255.255.255.0
vrrp 60 ip 192.168.60.1
vrrp 60 priority 110
interface GigabitEthernet0/2
no ip address
shutdown
duplex auto
speed auto
media-type rj45
interface GigabitEthernet0/3
no ip address
shutdown
duplex auto
speed auto

Terminal
Switch7 x vIOS-Jakarta x Fortinet1 x
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
Router#ping 10.10.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.1, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router#ping 10.10.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Router#

```

Gambar 72: Bukti (1) Konfigurasi IP subinterface, (2) Status VRRP, (3) Detail konfigurasi, dan (4) Ping ke FortiGate dari Cisco Router.

3. **Konfigurasi MikroTik Router Jakarta.** VLAN *interface* untuk VLAN 10, 20, dan 60 berhasil dikonfigurasi dengan IP fisik sesuai tabel pengalamatan. MikroTik berperan sebagai *master* VRRP untuk VLAN 20. DHCP *relay* dan koneksi ke FortiGate Jakarta berfungsi dengan baik. Rangkaian gambar ini membuktikan daftar IP Address, konfigurasi VRRP, penunjukan DHCP Relay, tabel *routing* aktif, serta keberhasilan ping menuju gateway FortiGate.


```

Mikrotik-Jak... > /ip address add address=10.10.10.2/30 interface=ether1
input does not match any value of interface
Mikrotik-Jak... > /ip address add address=10.10.10.2/30 interface=ether1
Mikrotik-Jak... > /ip address add address=10.10.10.2/30 interface=ether1
Mikrotik-Jak... > /ip address remove [find interface=vrrp10]
Mikrotik-Jak... > /ip address remove [find interface=vrrp20]
Mikrotik-Jak... > /ip address remove [find interface=vrrp60]
Mikrotik-Jak... > /ip address add address=192.168.10.1/32 interface=vrrp10
Mikrotik-Jak... > /ip address add address=192.168.20.1/32 interface=vrrp20
Mikrotik-Jak... > /ip address add address=192.168.60.1/32 interface=vrrp60
Mikrotik-Jak... > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 192.168.10.3/24 192.168.10.0 vlan10-finance
1 192.168.20.3/24 192.168.20.0 vlan20-it
2 192.168.60.3/24 192.168.60.0 vlan60-ubuntu-server
3 10.10.10.1/30 10.10.10.0 ether1
Mikrotik-Jak... > /ip dhcp-relay print
Flags: X - disabled, I - invalid, R - running, M - master, B - backup
# NAME INTERFACE MAC-ADDRESS VRRP PRI INTERVAL
0 RM vrrp10 vlan10-fin... 00:00:SE:00:01:0A 10 90 1s 3 ipw4
1 RM vrrp20 vlan20-it... 00:00:SE:00:01:14 20 120 1s 3 ipw4
2 RM vrrp60 vlan60-ubu... 00:00:SE:00:01:3C 60 90 1s 3 ipw4
Mikrotik-Jak... > /ip dhcp-relay print
Flags: X - disabled, I - invalid
# NAME INTERFACE DHCP-SERVER LOCAL-ADDRESS
0 relay-vlan10 vlan10-finance 192.168.60.10 192.168.10.3
1 relay-vlan20 vlan20-it 192.168.60.10 192.168.20.3
Mikrotik-Jak... > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 A S 0.0.0.0/0 10.10.10.1 1
1 ADC 10.10.10.0/30 10.10.10.2 ether1 0
2 ADC 192.168.10.0/24 192.168.10.3 vlan10-finance 0
3 ADC 192.168.10.1/32 192.168.10.1 vrrp10 0
4 ADC 192.168.20.0/24 192.168.20.3 vlan20-it 0
5 ADC 192.168.20.1/32 192.168.20.1 vrrp20 0
6 ADC 192.168.60.0/24 192.168.60.3 vlan60-ubuntu-s... 0
7 ADC 192.168.60.1/32 192.168.60.1 vrrp60 0
Mikrotik-Jak... >
Mikrotik-Jak... Fortinet1 x
18 10.10.10.1
sent=19 received=0 packet-loss=100%
timeout
Mikrotik-Jak... > ping 10.10.10.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.10.1 56 255 0ms
1 10.10.10.1 56 255 0ms
2 10.10.10.1 56 255 0ms
3 10.10.10.1 56 255 0ms
4 10.10.10.1 56 255 0ms
5 10.10.10.1 56 255 0ms
6 10.10.10.1 56 255 0ms
7 10.10.10.1 56 255 0ms
sent=8 received=8 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms
Mikrotik-Jak... >

```

Gambar 73: Bukti (1-3) IP, VRRP, dan DHCP Relay, serta (4-5) Tabel routing dan hasil ping Mikrotik Jakarta.

4. **Konfigurasi Ubuntu Server Jakarta.** Ubuntu Server dikonfigurasi dengan IP statis pada VLAN 60 dan *gateway* mengarah ke IP virtual VRRP. ISC-DHCP Server berhasil melayani permintaan IP dari klien VLAN 10 dan VLAN 20 melalui DHCP *relay*. Gambar di bawah memvalidasi konfigurasi `ip a`, rute lokal (`ip route`), isi file konfigurasi `dhcpd.conf` yang memuat *pool* tiap VLAN, serta konektivitas internet melalui ping ke 8.8.8.8.

```

root@kum:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 50:9f:b8:02:37:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.60.10/24 brd 192.168.60.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::529f:b8ff:fe02:3700/64 scope link
        valid_lft forever preferred_lft forever
root@kum:~#

GNU nano 4.8 dhcpd.conf
#}

#shared-network 224-29 {
#    subnet 10.17.224.0 netmask 255.255.255.0 {
#        option routers rtr-224.example.org;
#    }
#    subnet 10.0.29.0 netmask 255.255.255.0 {
#        option routers rtr-29.example.org;
#    }
#    pool {
#        allow members of "foo";
#        range 10.17.224.10 10.17.224.250;
#    }
#    pool {
#        deny members of "foo";
#        range 10.0.29.10 10.0.29.230;
#    }
#}

authoritative;
default-lease-time 600;
max-lease-time 7200;
option domain-name-servers 8.8.8.8, 1.1.1.1;

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.100 192.168.10.200;
    option routers 192.168.10.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.10.255;
}

subnet 192.168.20.0 netmask 255.255.255.0 {
    range 192.168.20.100 192.168.20.200;
    option routers 192.168.20.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.20.255;
}

subnet 192.168.60.0 netmask 255.255.255.0 {
    option routers 192.168.60.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.60.255;
}

root@kum:/etc/default# ip route
default via 192.168.60.1 dev eth0 proto static
192.168.60.0/24 dev eth0 proto kernel scope link src 192.168.60.10
root@kum:~#

root@kum:/etc/default# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
^C
--- 8.8.8.8 ping statistics ---
0 packets transmitted, 0 received, 100% packet loss, time 8197ms
root@kum:/etc/default#

```

Gambar 74: Bukti (1) IP Address Server, (2) Tabel rute, (3) Konfigurasi ISC-DHCP Server, dan (4) Ping internet dari Ubuntu Server.

5. **Konfigurasi FortiGate Jakarta.** Interface menuju Cisco Router, MikroTik Router, dan MikroTik ISP berhasil dikonfigurasi beserta *routing* dan *firewall policy* yang mengizinkan akses internet. OSPF over GRE menuju Surabaya juga berhasil dibentuk. Gambar berikut merangkum daftar *physical interface*, tabel *routing* keseluruhan, aturan *firewall policy*, ping internet & tunnel Surabaya, status OSPF *neighbor* (Full), dan rute OSPF yang diterima.


```

Terminal
Mikrotik-Jak... x Switch-Jakarta x Mikrotik-ISP x
0 10.0.13.2 tim
1 10.0.13.2 tim
2 10.0.13.1 84 64 988ms hos
sent=3 received=0 packet-loss=100%

[admin@MikroTik] >
[admin@MikroTik] > /ip address remove 2
[admin@MikroTik] > /ip firewall nat remove 1
[admin@MikroTik] > /ip firewall nat remove 2
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; Ke Fortigate Jakarta 10.0.12.1/30 10.0.12.0 ether2
1 ;; Ke Fortigate Surabaya 10.0.13.1/30 10.0.13.0 ether3
2 D 10.0.137.152/24 10.0.137.0 ether1
[admin@MikroTik] > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY
0 ADS 0.0.0.0/0 10.0.137.1 10.0.137.1
1 ADC 10.0.12.0/30 10.0.12.1 ether2
2 ADC 10.0.13.0/30 10.0.13.1 ether3
3 ADC 10.0.137.0/24 10.0.137.152 ether1
[admin@MikroTik] > /ip firewall nat remove 1
no such item (4)
[admin@MikroTik] > /ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1
[admin@MikroTik] > ping 8.8.8.8
SEQ HOST SIZE TTL TIME STA
0 8.8.8.8 56 112 24ms
1 8.8.8.8 56 112 23ms
2 8.8.8.8 56 112 23ms
sent=3 received=3 packet-loss=0% min-rtt=23ms avg-rtt=23ms
max-rtt=24ms

[admin@MikroTik] > ping 10.0.12.2
SEQ HOST SIZE TTL TIME STA
0 10.0.12.2 56 255 0ms

Mikrotik-ISP x
[admin@MikroTik] /ip firewall nat /ping 10.0.12.2 count=5
SEQ HOST SIZE TTL TIME STATUS
0 10.0.12.2 56 255 0ms
1 10.0.12.2 56 255 0ms
2 10.0.12.2 56 255 0ms
3 10.0.12.2 56 255 0ms
4 10.0.12.2 56 255 0ms
sent=5 received=5 packet-loss=0% min-rtt=0ms avg-rtt=0ms
max-rtt=0ms

[admin@MikroTik] /ip firewall nat /ping 10.0.13.2 count=5
SEQ HOST SIZE TTL TIME STATUS
0 10.0.13.2 56 255 0ms
1 10.0.13.2 56 255 0ms
2 10.0.13.2 56 255 0ms
3 10.0.13.2 56 255 0ms
4 10.0.13.2 56 255 0ms
sent=5 received=5 packet-loss=0% min-rtt=0ms avg-rtt=0ms
max-rtt=0ms

[admin@MikroTik] /ip firewall nat /ping 8.8.8.8 count=5
SEQ HOST SIZE TTL TIME STATUS
0 8.8.8.8 56 112 24ms
1 8.8.8.8 56 112 23ms
2 8.8.8.8 56 112 23ms
3 8.8.8.8 56 112 23ms
4 8.8.8.8 56 112 24ms
sent=5 received=5 packet-loss=0% min-rtt=23ms avg-rtt=23ms
max-rtt=24ms

Terminal
Mikrotik-Jak... x Switch-Jakarta x Mikrotik-ISP x
max-rtt=0ms

[admin@MikroTik] > ping 10.0.13.2
SEQ HOST SIZE TTL TIME STA
0 10.0.13.2 tim
1 10.0.13.2 tim
2 10.0.13.1 84 64 985ms hos
sent=3 received=0 packet-loss=100%

[admin@MikroTik] > ping 10.0.13.2
SEQ HOST SIZE TTL TIME STA
0 10.0.13.2 tim
1 10.0.13.2 tim
2 10.0.13.2 tim
3 10.0.13.2 tim
4 10.0.13.2 tim
5 10.0.13.2 tim
sent=6 received=0 packet-loss=100%

[admin@MikroTik] /ip firewall nat /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 D 10.0.137.202/24 10.0.137.0 ether1
1 ;; Ke Fortigate Jakarta 10.0.12.1/30 10.0.12.0 ether2
[admin@MikroTik] /ip firewall nat /ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1 terface=ether1 action=masquerade
1 chain=srcnat action=masquerade out-interface=ether1
[admin@MikroTik] /ip firewall nat /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.0.137.1 1
1 ADC 10.0.12.0/30 10.0.12.1 ether2 0
2 ADC 10.0.13.0/30 10.0.13.1 ether3 0
3 ADC 10.0.137.0/24 10.0.137.202 ether1 0

```

Gambar 76: Bukti pengalamatan, NAT, dan pengujian jalur WAN lintas FortiGate pada MikroTik ISP.

7. **Konfigurasi Switch dan MikroTik Surabaya.** VLAN 30 dan VLAN 40 aktif di Switch Surabaya. MikroTik Surabaya berperan sebagai *gateway* dengan DHCP lokal untuk VLAN 30. Gambar berikut menunjukkan konfigurasi VLAN dan *trunk* pada switch, dilanjutkan dengan IP address, pengaturan DHCP Server & Pool, tabel *routing* MikroTik, serta bukti perolehan IP DHCP dan ping internet dari sisi VPCS klien Surabaya.

```

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x Fortinet2 x
Switch#show vlan brief
VLAN Name                Status    Ports
----
1  default                active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
10  VLAN0010                active
20  VLAN0020                active
30  sales                   active    Gi0/1
40  operations               act/unsu  Gi0/2, Gi0/3
1002 fddi-default            act/unsu
1003 token-ring-default      act/unsu
1004 fddinet-default        act/unsu
1005 trnet-default          act/unsu
Switch#

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x
switch#show interfaces trunk
Port      Mode      Encapsulation  Status    Native
-----
Gi0/0     on        802.1q          trunking  1
Gi0/1     on        802.1q          trunking  1
Gi0/2     on        802.1q          trunking  1
Gi0/3     on        802.1q          trunking  1
Vlans allowed on trunk
Gi0/0     30,40
Gi0/1     30,40
Gi0/2     30,40
Gi0/3     30,40
Vlans allowed and active in management domain
Gi0/0     30,40
Gi0/1     30,40
Gi0/2     30,40
Gi0/3     30,40
Vlans in spanning tree forwarding state and not pruned
Gi0/0     30,40
Gi0/1     30,40
Gi0/2     30,40
Gi0/3     30,40
Switch#

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
#  ADDRESS          NETWORK    INTERFACE
0  192.168.30.1/24    192.168.30.0  vlan30-sales
1  192.168.40.1/24    192.168.40.0  vlan40-operation
2  10.10.200.2/30     10.10.200.0   ether1
[admin@MikroTik] > /ip dhcp-server print
Flags: D - dynamic, X - disabled, I - invalid
#  NA IN RELAY  ADDRESS-POOL  LEASE-TIME  ADD
0  dh vl       dhcp_pool0    10m
[admin@MikroTik] > /ip pool print
#  NAME    RANGES
0  dhcp_pool0  192.168.30.2-192.168.30.254
[admin@MikroTik] > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#  DST-ADDRESS  PREF-SRC
0  A S  0.0.0.0/0
1  ADC  10.10.200.0/30  10.10.200.2
2  ADC  192.168.30.0/24  192.168.30.1
3  ADC  192.168.40.0/24  192.168.40.1
[admin@MikroTik] >

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x
VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 192.168.30.254/24
GATEWAY   : 192.168.30.1
DNS       : 8.8.8.8
DHCP SERVER : 192.168.30.1
DHCP LEASE : 595, 600/300/525
MAC       : 00:50:79:66:68:35
PORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x Fortinet2 x
[admin@MikroTik] > ping 10.10.200.1
ping: send: no route to host (destination host unreachable)
10.10.200.1 icmp_seq=3 timeout
*192.168.30.1 icmp_seq=4 ttl=64 time=999.287 ms (ICMP type:3, c
[admin@MikroTik] > ping 10.10.200.1
ping: send: no route to host (destination host unreachable)
10.10.200.1 icmp_seq=5 timeout
^C
VPCS> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=109 time=27.389 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=109 time=26.152 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=109 time=26.021 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=109 time=27.141 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=109 time=26.462 ms
^C

```

Gambar 77: Bukti (1-2) Konfigurasi Switch Surabaya, (3-6) Parameter MikroTik SBY, dan (7-8) Hasil IP DHCP & Ping klien.

8. **Konfigurasi FortiGate Surabaya.** Konfigurasi interface, *routing*, *firewall policy*, NAT, dan GRE Tunnel berhasil diterapkan layaknya HQ Jakarta. Gambar di bawah ini membuktikan status *physical interface*, tabel *routing* lengkap, *firewall policy* lokal, uji ping ke 8.8.8.8 dan IP Tunnel Jakarta, serta status OSPF *neighbor* yang mencapai tahap *Full* diikuti dengan penerimaan rute OSPF.



Gambar 78: Bukti (1-3) Interface, Routing, & Policy FortiGate Surabaya, serta (4-7) Pengujian GRE Tunnel dan status OSPF.

9. **Konfigurasi GRE Tunnel dan OSPF over GRE.** Tunnel antar-FortiGate sukses mengalirkan protokol OSPF. Gambar berikut secara berurutan membuktikan keberhasilan ping pada IP WAN antar-FortiGate, ping melalui lorong IP Tunnel, status OSPF *neighbor*, tabel rute OSPF yang mendistribusikan jalur antarsite, serta pengujian akhir berupa keberhasilan ping lalu lintas data dari Klien Jakarta ke Klien Surabaya dan sebaliknya.

```

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x Fortinet2 x Fortinet1 x
FortiGate-VM64-KVM # Timeout
FortiGate-VM64-KVM login: admin
Password:
Welcome!
FortiGate-VM64-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.1 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=0.5 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=0.5 ms
64 bytes from 10.0.12.2: icmp_seq=3 ttl=254 time=0.7 ms
^C
--- 10.0.12.2 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.5/0.7/1.1 ms
FortiGate-VM64-KVM #

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x Fortinet2 x Fortinet1 x
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=0.5 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=0.5 ms
64 bytes from 10.0.12.2: icmp_seq=3 ttl=254 time=0.7 ms
^C
--- 10.0.12.2 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.5/0.7/1.1 ms
FortiGate-VM64-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=0.7 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=0.6 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=0.6 ms
64 bytes from 172.16.0.1: icmp_seq=3 ttl=255 time=0.5 ms
^C
--- 172.16.0.1 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.5/0.6/0.7 ms
FortiGate-VM64-KVM #

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x Fortinet2 x Fortinet1 x
FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID Pri State Dead Time Address Interface
1.1.1.1 1 Full/ - 00:00:38 172.16.0.1 GRE-SBY-JKT

FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
0 E2 192.168.10.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 00:36:36, [1/0]
0 E2 192.168.20.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 00:36:08, [1/0]
0 E2 192.168.60.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 00:35:53, [1/0]
FortiGate-VM64-KVM #

Terminal
Mikrotik-ISP x Mikrotik-Sur... x Switch-Sura... x VLAN 30 x Fortinet2 x Fortinet1 x
FortiGate-VM64-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=0.7 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=0.6 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=0.6 ms
64 bytes from 172.16.0.1: icmp_seq=3 ttl=255 time=0.5 ms
^C
--- 172.16.0.1 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.5/0.6/0.7 ms
FortiGate-VM64-KVM #

Terminal
Fortinet2 x Fortinet1 x VLAN 10 x VLAN 30 x
192.168.30.254 icmp_seq=3 timeout
192.168.30.254 icmp_seq=4 timeout
^C
VPCS> ping 192.168.30.254
84 bytes from 192.168.30.254 icmp_seq=1 ttl=60 time=6.182 ms
84 bytes from 192.168.30.254 icmp_seq=2 ttl=60 time=10.191 ms
84 bytes from 192.168.30.254 icmp_seq=3 ttl=60 time=4.716 ms
84 bytes from 192.168.30.254 icmp_seq=4 ttl=60 time=5.581 ms
84 bytes from 192.168.30.254 icmp_seq=5 ttl=60 time=6.210 ms
^C
VPCS>

Terminal
Fortinet2 x Fortinet1 x VLAN 10 x VLAN 30 x
192.168.10.100 icmp_seq=4 timeout
192.168.10.100 icmp_seq=5 timeout
^C
VPCS> ping 192.168.10.100
84 bytes from 192.168.10.100 icmp_seq=1 ttl=60 time=5.501 ms
84 bytes from 192.168.10.100 icmp_seq=2 ttl=60 time=5.893 ms
84 bytes from 192.168.10.100 icmp_seq=3 ttl=60 time=4.407 ms
84 bytes from 192.168.10.100 icmp_seq=4 ttl=60 time=3.898 ms
84 bytes from 192.168.10.100 icmp_seq=5 ttl=60 time=4.327 ms
^C
VPCS>

```

Gambar 79: Bukti validasi tunnel WAN/GRE, status OSPF terdistribusi, dan konektivitas Site-to-Site.

10. **Pengujian Akhir.** Pengujian menyeluruh memvalidasi keberhasilan topologi. Rangkaian gambar terakhir ini menampilkan bukti perolehan IP DHCP oleh Klien Jakarta (VLAN 10) dan Klien Surabaya (VLAN 20), akses internet dari kedua sisi site, uji konektivitas antar-site lintas VLAN, aksesibilitas Web Server Jakarta dari Surabaya, serta kondisi stabil dari tabel *routing* OSPF.

```

Terminal
Fortinet2 x Fortinet1 x VLAN 10 x VLAN 30 x
VPCS> ping 192.168.30.254
84 bytes from 192.168.30.254 icmp_seq=1 ttl=60 time=6.192 ms
84 bytes from 192.168.30.254 icmp_seq=2 ttl=60 time=10.191 ms
84 bytes from 192.168.30.254 icmp_seq=3 ttl=60 time=4.716 ms
84 bytes from 192.168.30.254 icmp_seq=4 ttl=60 time=5.581 ms
84 bytes from 192.168.30.254 icmp_seq=5 ttl=60 time=6.210 ms
^C
VPCS> ip dhcp
DDORA IP 192.168.10.101/24 GW 192.168.10.1
VPCS>

Terminal
Fortinet2 x Fortinet1 x VLAN 10 x VLAN 30 x
84 bytes from 192.168.30.254 icmp_seq=5 ttl=60 time=6.210 ms
^C
VPCS> ip dhcp
DDORA IP 192.168.10.101/24 GW 192.168.10.1
VPCS> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=109 time=25.621 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=109 time=25.316 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=109 time=25.420 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=109 time=26.565 ms
^C
VPCS>

Terminal
Fortinet2 x Fortinet1 x VLAN 10 x VLAN 30 x
84 bytes from 8.8.8.8 icmp_seq=1 ttl=109 time=25.621 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=109 time=25.316 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=109 time=25.420 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=109 time=26.565 ms
^C
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=60 time=7.183 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=60 time=3.932 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=60 time=3.503 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=60 time=6.106 ms
^C
VPCS>

Terminal
Fortinet2 x Fortinet1 x VLAN 10 x VLAN 30 x
VPCS> ping 192.168.10.100
84 bytes from 192.168.10.100 icmp_seq=1 ttl=60 time=5.501 ms
84 bytes from 192.168.10.100 icmp_seq=2 ttl=60 time=5.893 ms
84 bytes from 192.168.10.100 icmp_seq=3 ttl=60 time=4.407 ms
84 bytes from 192.168.10.100 icmp_seq=4 ttl=60 time=3.898 ms
84 bytes from 192.168.10.100 icmp_seq=5 ttl=60 time=4.327 ms
^C
VPCS> ip dhcp
DDORA IP 192.168.30.254/24 GW 192.168.30.1
VPCS>

Terminal
Fortinet2 x Fortinet1 x VLAN 10 x VLAN 30 x
84 bytes from 192.168.10.100 icmp_seq=5 ttl=60 time=4.327 ms
^C
VPCS> ip dhcp
DDORA IP 192.168.30.254/24 GW 192.168.30.1
VPCS> ping 8.8.8.8
84 bytes from 8.8.8.8 icmp_seq=1 ttl=109 time=26.021 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=109 time=26.054 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=109 time=25.598 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=109 time=24.944 ms
^C
VPCS>

Terminal
FortiGate-VM64-KVM login: admin
Password:
Welcome!

FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
0 E2 192.168.10.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:13:17, [1/0]
0 E2 192.168.20.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:12:49, [1/0]
0 E2 192.168.60.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:12:34, [1/0]

FortiGate-VM64-KVM #

```

Gambar 80: Bukti pengujian End-to-End: DHCP, Ping Internet, inter-VLAN antarsite, akses Web Server, dan rute OSPF final.

Analisis Jalur Traffic Jakarta - Surabaya: Lalu lintas data dari jaringan Jakarta menuju Surabaya berawal dari PC Client yang mengirimkan paket ke Cisco Router (berfungsi sebagai gateway redundan via VRRP). Cisco Router kemudian meneruskan paket ke FortiGate Jakarta. Mengingat tujuannya adalah site Surabaya, FortiGate Jakarta mengenkapsulasi paket tersebut ke dalam GRE Tunnel yang rutanya diatur secara dinamis oleh OSPF. Paket terenkapsulasi ini melintasi jaringan MikroTik ISP hingga tiba di FortiGate Surabaya. Di sana, FortiGate membuka enkapsulasi GRE dan meneruskan paket asli ke MikroTik Surabaya, yang pada akhirnya menyalurkan paket tersebut melalui Switch ke PC Client tujuan.

Secara keseluruhan, sepuluh bagian tugas modul tersebut berhasil mengimplementasikan jaringan *enterprise* dua *site* dengan mekanisme *dual gateway* menggunakan VRRP, DHCP terpusat melalui ISC-DHCP dan DHCP *relay*, keamanan dan NAT melalui FortiGate, serta konektivitas antarsite melalui GRE Tunnel yang menjalankan OSPF di atasnya. Penjelasan rinci mengenai konfigurasi setiap perangkat beserta bukti pendukung tercantum pada berkas *readme* terpisah.

Secara keseluruhan, sepuluh bagian tugas modul tersebut berhasil mengimplementasikan jaringan *enterprise* dua *site* dengan mekanisme *dual gateway* menggunakan VRRP, DHCP terpusat melalui ISC-DHCP dan DHCP *relay*, keamanan dan NAT melalui FortiGate, serta konektivitas antarsite melalui GRE Tunnel yang menjalankan OSPF di atasnya. Penjelasan rinci mengenai konfigurasi setiap perangkat beserta bukti pendukung tercantum pada berkas *readme* terpisah.

4 Kesimpulan

Praktikum Modul 5 berhasil mengimplementasikan jaringan *multivendor* yang mengintegrasikan perangkat Cisco, Huawei, dan MikroTik menggunakan protokol routing dinamis OSPF dalam satu topologi terpadu.

Pada Praktikum 1, Cisco Switch berhasil dikonfigurasi dengan empat VLAN aktif, yaitu VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing). *Access port* ditetapkan pada setiap interface yang mengarah ke PC, dan *trunk port* dikonfigurasi pada interface menuju Cisco Router dengan enkapsulasi 802.1Q sehingga keempat VLAN dapat melewati satu link fisik sekaligus.

Pada Praktikum 2, Cisco Router berhasil dikonfigurasi sebagai *gateway* untuk keempat VLAN menggunakan metode ROAS melalui *subinterface* berbasis 802.1Q. DHCP server aktif untuk VLAN 10 dan VLAN 20 sehingga PC pada kedua VLAN tersebut memperoleh IP secara otomatis, sedangkan PC VLAN 30 dan VLAN 40 menggunakan IP statis. Seluruh PC berhasil melakukan ping ke *gateway* masing-masing dan ping antar-VLAN terbukti berfungsi melalui Cisco Router.

Pada Praktikum 3, IP address berhasil dikonfigurasi pada keempat link antarrouter. LAN Huawei dengan jaringan 192.168.50.0/24 aktif dan PC LAN Huawei berhasil melakukan ping ke *gateway*. Setiap router berhasil melakukan ping langsung ke router tetangga, membuktikan konektivitas layer 3 pada seluruh link telah berfungsi sebelum OSPF dikonfigurasi.

Pada Praktikum 4, OSPF *multivendor* berhasil dikonfigurasi pada Cisco Router (*router-id* 1.1.1.1), Huawei Router (*router-id* 2.2.2.2), dan MikroTik (*router-id* 3.3.3.3). Nilai *cost* 10 ditetapkan pada dua link Cisco ke Huawei sebagai jalur utama, sedangkan nilai *cost* 100 ditetapkan pada link melalui MikroTik sebagai jalur cadangan. Ketiga router berhasil membentuk *neighbor* OSPF dan saling mempertukarkan informasi routing. PC di seluruh VLAN Cisco berhasil melakukan ping ke PC LAN Huawei dan sebaliknya, membuktikan bahwa OSPF berhasil mendistribusikan rute ke seluruh jaringan lintas vendor.

Pada Praktikum 5, mekanisme *failover* OSPF berhasil diverifikasi melalui empat skenario. Pada kondisi normal, trafik melewati jalur langsung Cisco ke Huawei. Saat satu link utama dinonaktifkan, OSPF secara otomatis mengalihkan trafik ke link utama yang tersisa. Saat kedua link utama dinonaktifkan, OSPF mengalihkan trafik ke jalur cadangan melalui MikroTik dengan urutan hop: PC VLAN, Cisco Router, MikroTik, Huawei Router, dan LAN Huawei. Ketika kedua link utama diaktifkan kembali, OSPF secara otomatis mengembalikan trafik ke jalur utama. Hal ini membuktikan bahwa mekanisme pemilihan jalur OSPF berdasarkan nilai *cost* berfungsi dengan benar sebagai solusi *failover* otomatis pada jaringan *multivendor*.

5 Lampiran



Gambar 81: Dokumentasi Sesi Praktikum